

Week 03. Variance. Bi-variate data. Correlation and Causation.

Part 2: Statistics and Econometrics. MK68-2023/24

Eduard Bukin

eduard.bukin@agrar.uni-giessen.de

Justus Liebig University Giessen

Institute of Agricultural Policy and Market Research

Content

- Homework for the Christmas break
- Bi-variate data
- Scatter plot
- Trend line
- Correlation
- Correlation and Causation
- Exploratory analysis example: rice yields and fertilizers application
- Cause, Effect, Potential Outcome and Ceteris Paribus
- At home: New insights on the HIV pandemic
- Other homework
- References

Homework for the Christmas break

Watch these videos in exact order:

1. [Econometrics: The Path from Cause to Effect](#)
2. [Ceteris Paribus: Public vs. Private University](#)
3. [Selection Bias: Will You Make More Going to a Private University?](#)
4. [Randomized Trials: The Ideal Weapon](#)

More homework in the end of slides.

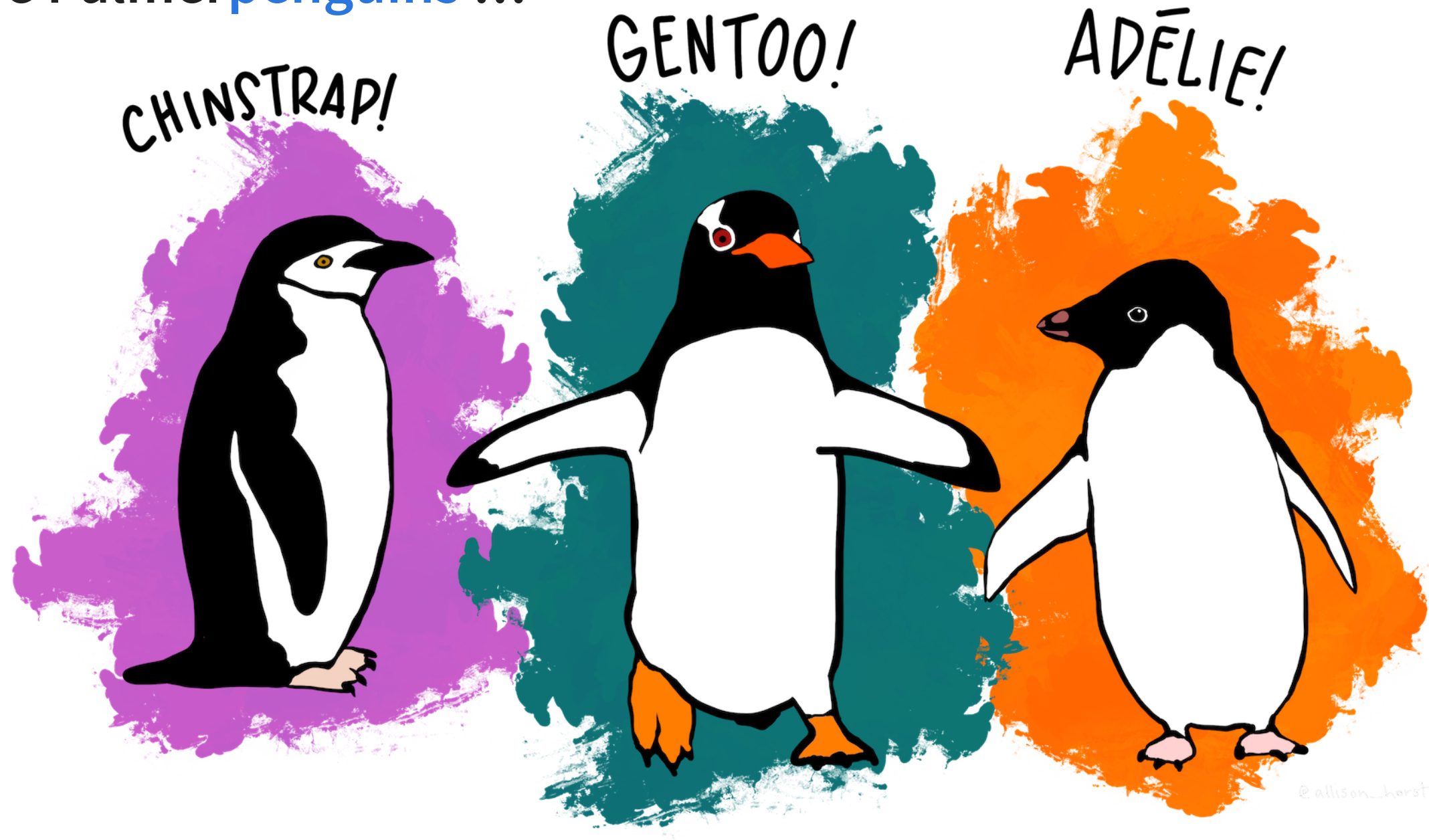
Bi-variate data

Data on two variables:

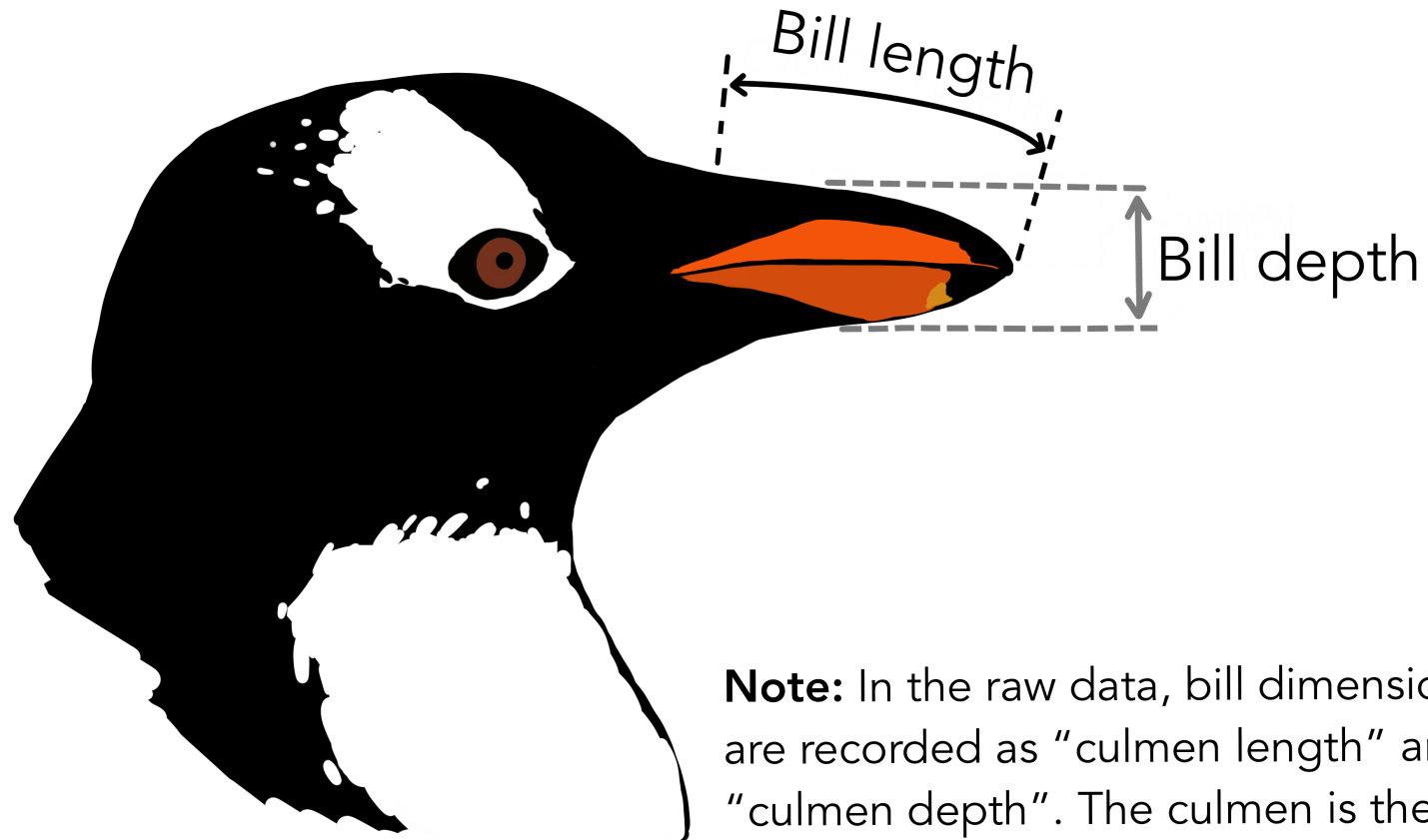
- Where each variable is paired with another.

Example of Bi-variate data... (1)

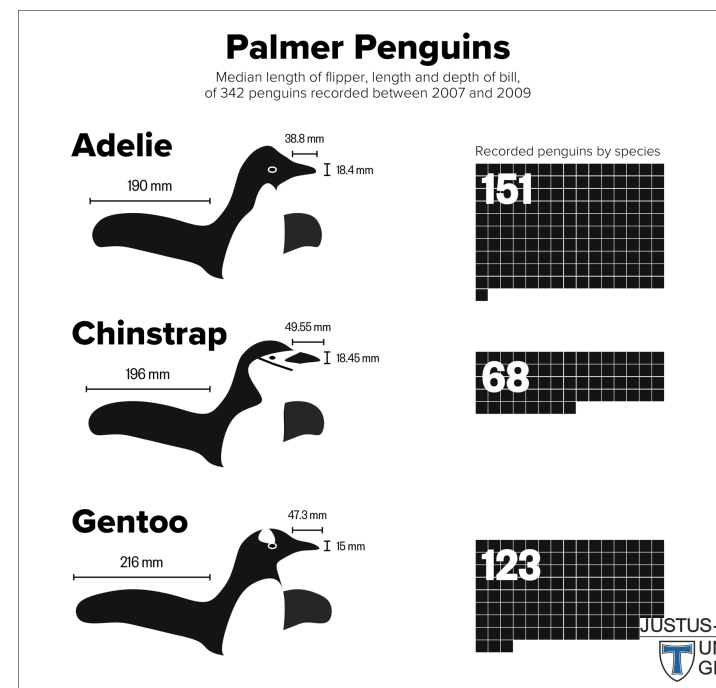
are Palmer **penguins** !!!



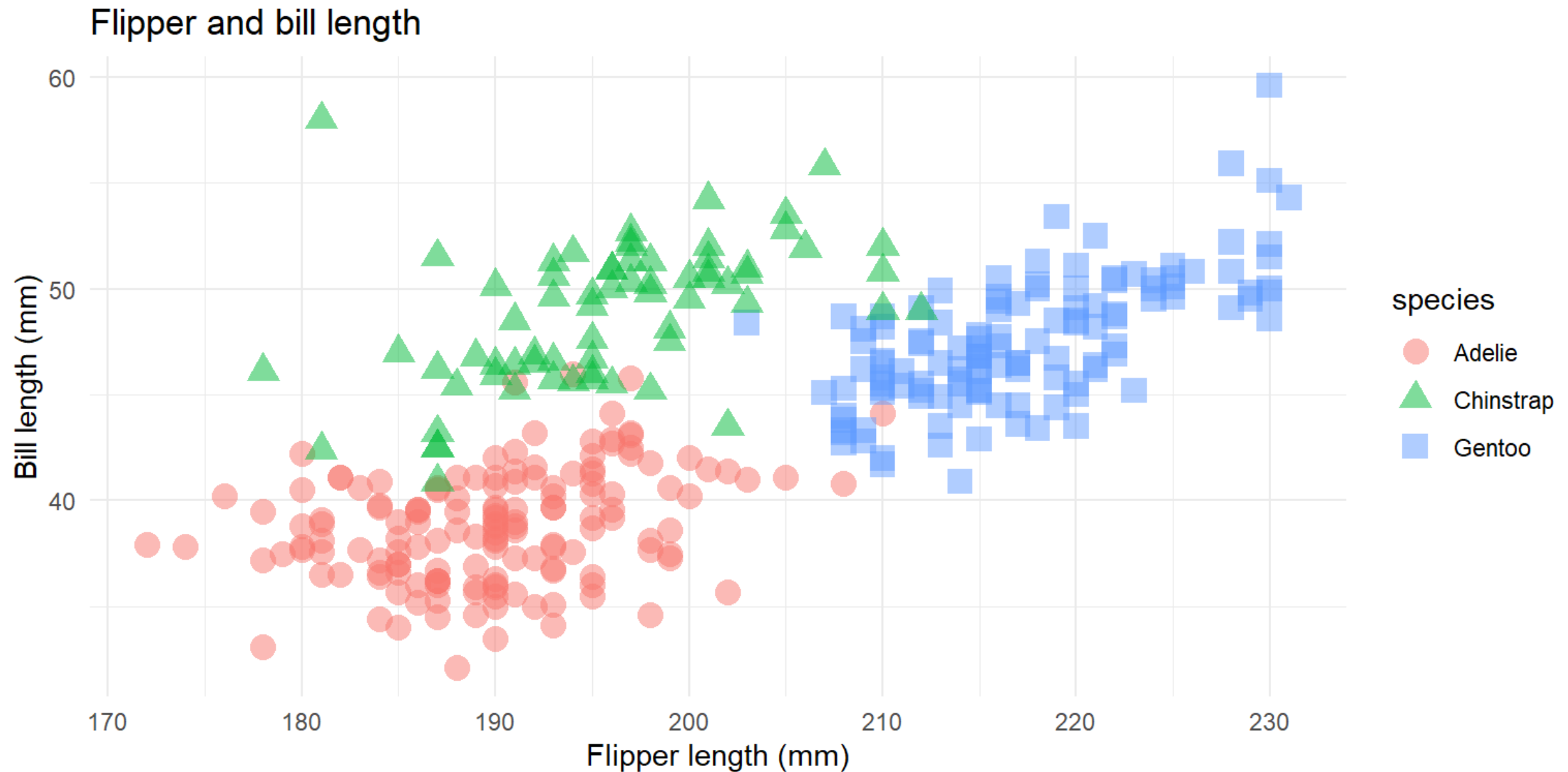
Example of Bi-viriate data... (2)



Note: In the raw data, bill dimensions are recorded as "culmen length" and "culmen depth". The culmen is the dorsal ridge atop the bill.



Building a scatter plot with penguins



Scatter plot

Type of a plot that shows the relationship between two numeric variables.

Convention:

- x is the horizontal axis and y is the vertical;
- x changes first and then y follows;
- if any causality is implied, x causes y ;

More information:

- Khan Academy. [Unit: Exploring bivariate numerical data. Introduction to scatter plots](#)

Good scatter plot is self explaining

It contains:

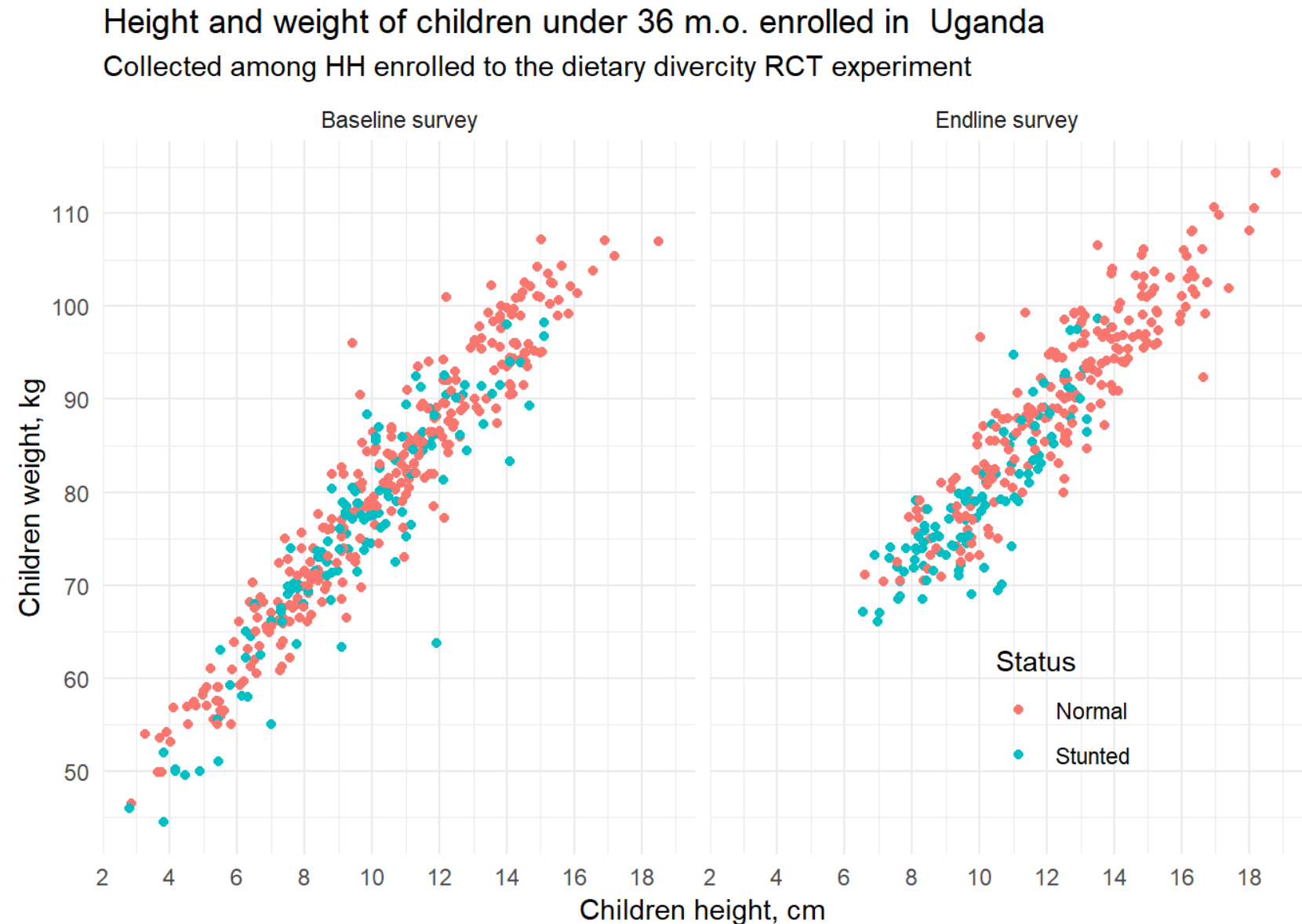
Title and sources

Axis labels, units

Points

Legends

Temporal dimension



Source: Jordan I, Röhlig A, Glas MG, Waswa LM, Mugisha J, Krawinkel MB, Nuppenau E-A. Dietary Diversity of Women across Agricultural Seasons in the Kapchorwa District, Uganda: Results from a Cohort Study. *Foods*. 2022; 11(3):344. <https://doi.org/10.3390/foods11030344>

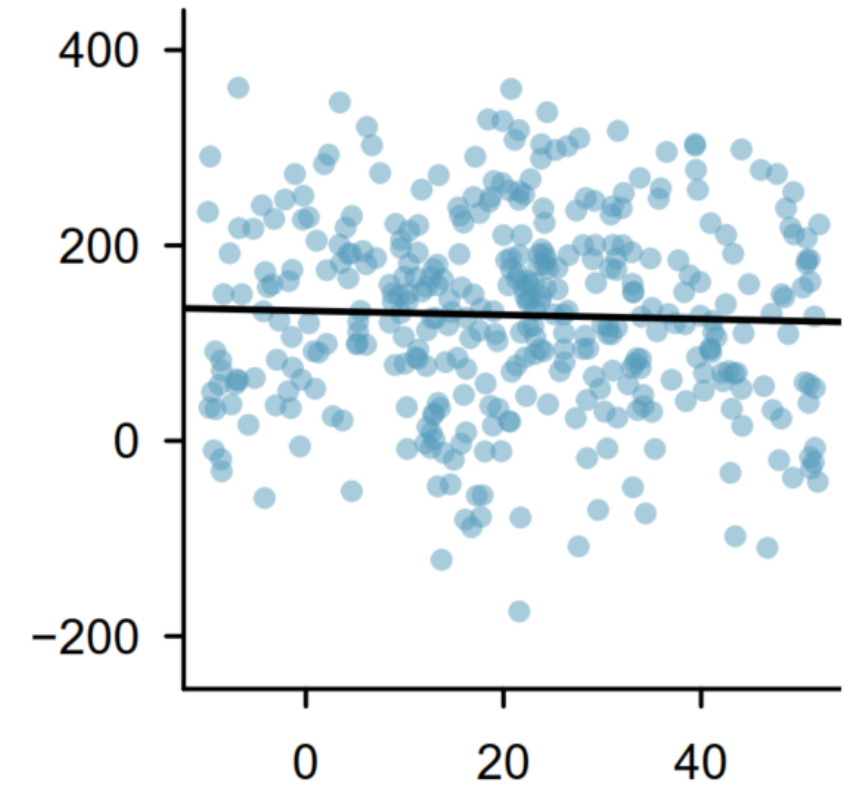
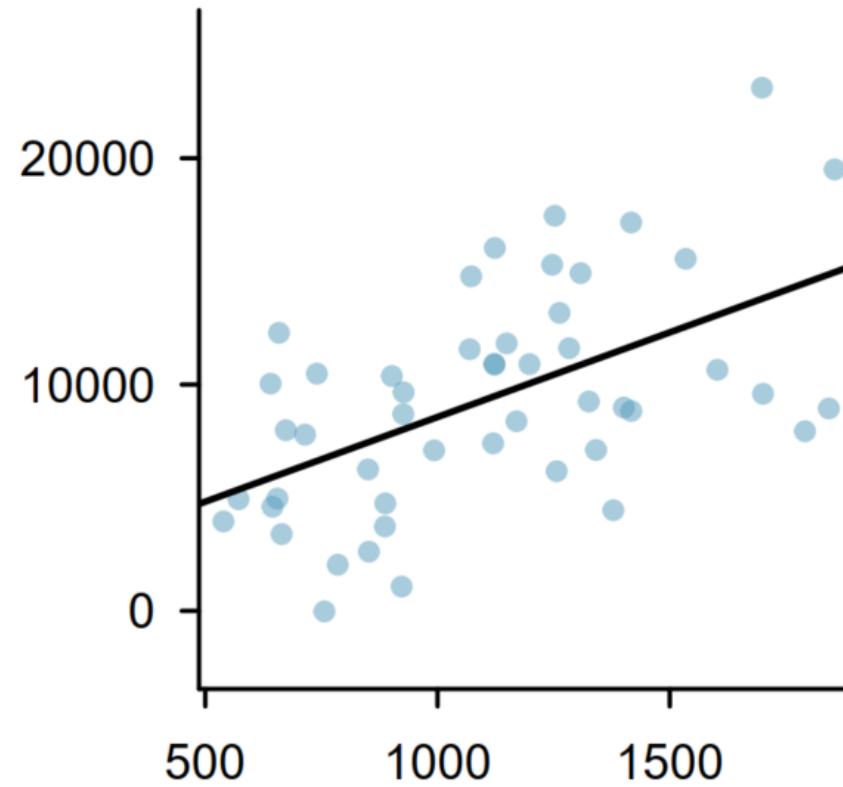
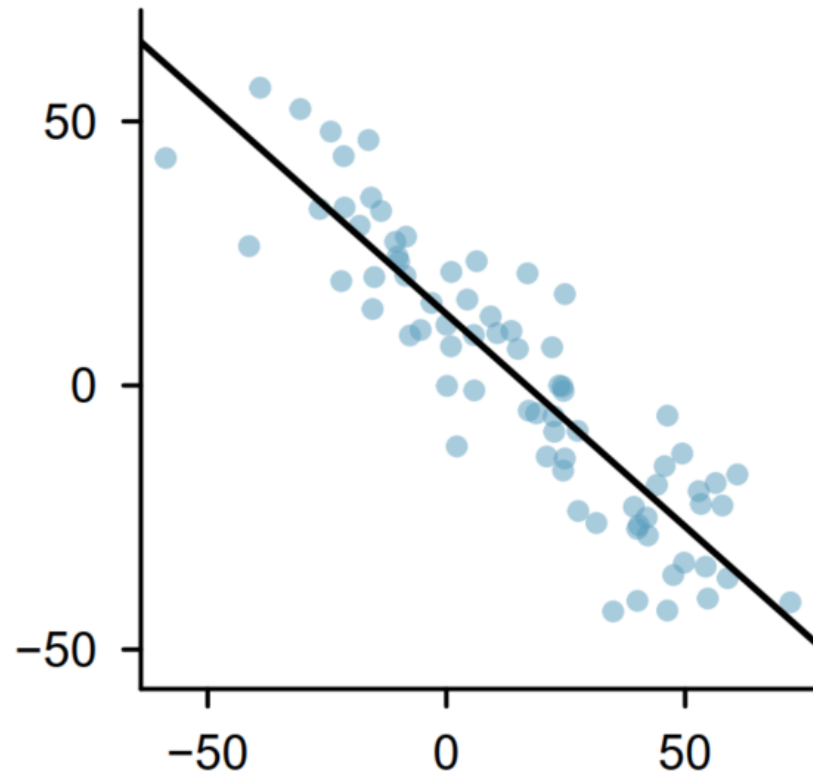
Trend line

Is the line that best outlines a linear trend in the relationship between x and y .

More information:

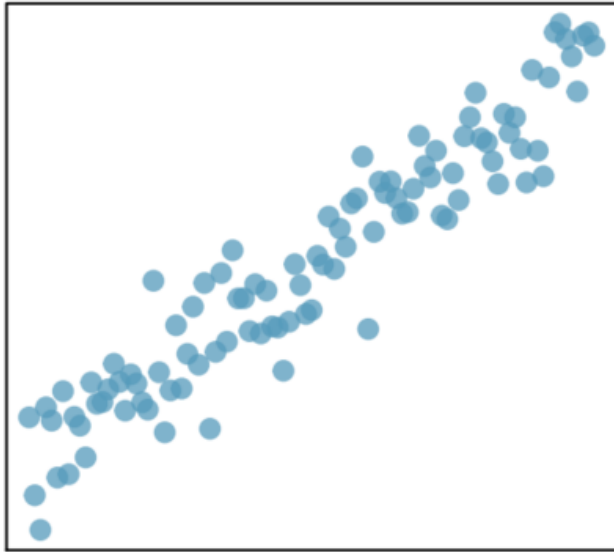
- Khan Academy. [Unit: Exploring bivariate numerical data. Introduction to trend lines](#)

Trend line examples

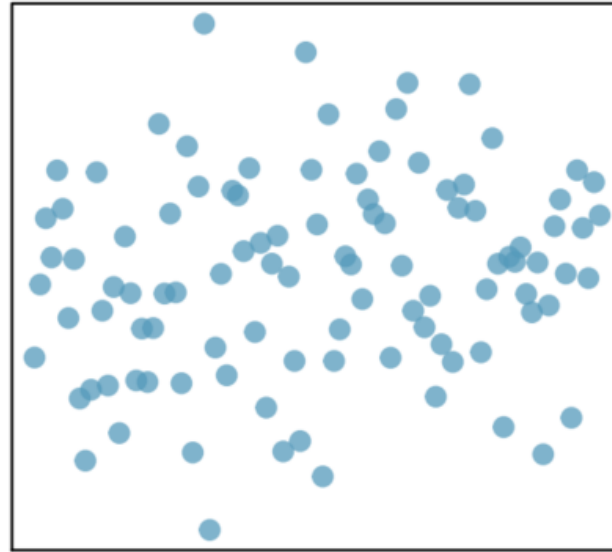


Examples: describe the scatter plots (1)

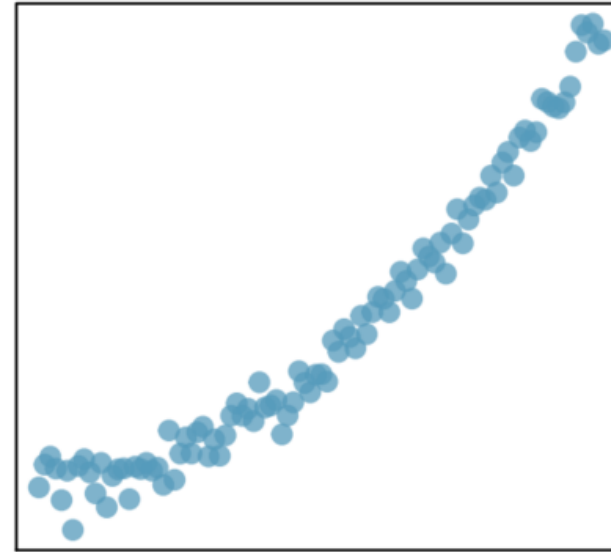
What are the associations between X and Y ? Are those positive/negative, linear/nonlinear? How but is the variance of Y given X ?



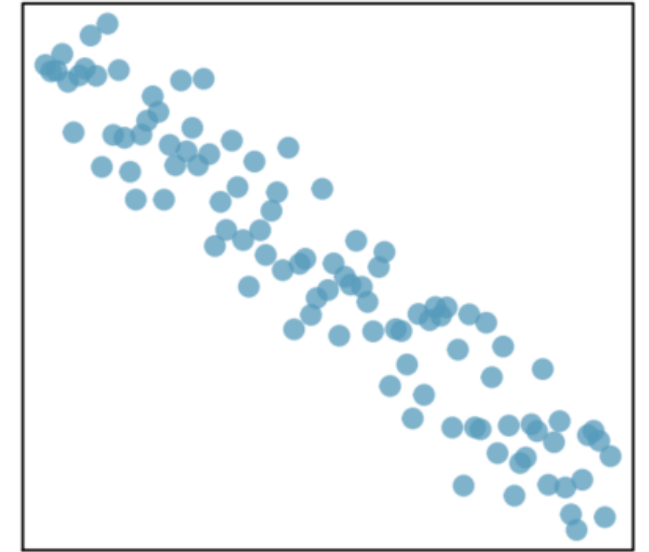
(1)



(2)



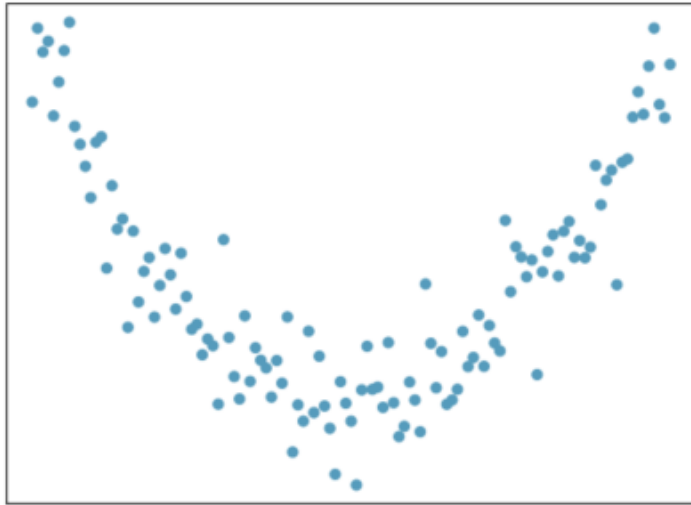
(3)



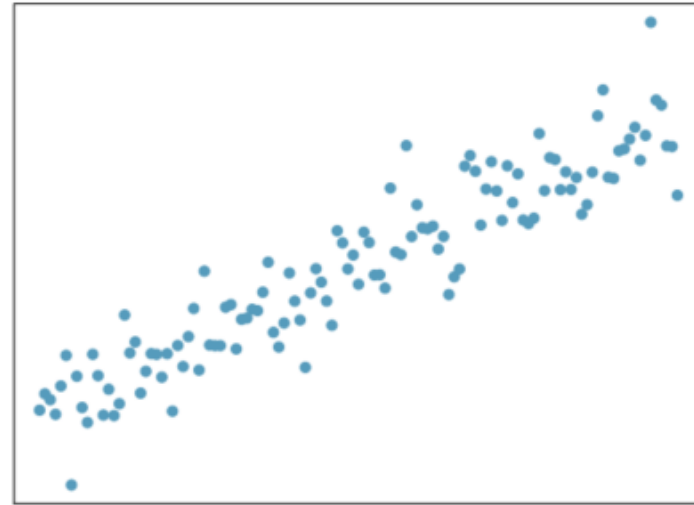
(4)

Examples: describe the scatter plots (2)

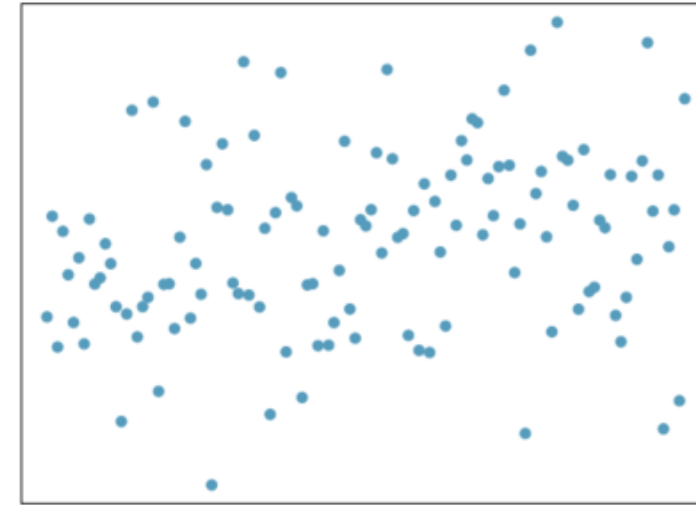
What are the associations between X and Y ? Are those positive/negative, linear/nonlinear? How but is the variance of Y given X ?



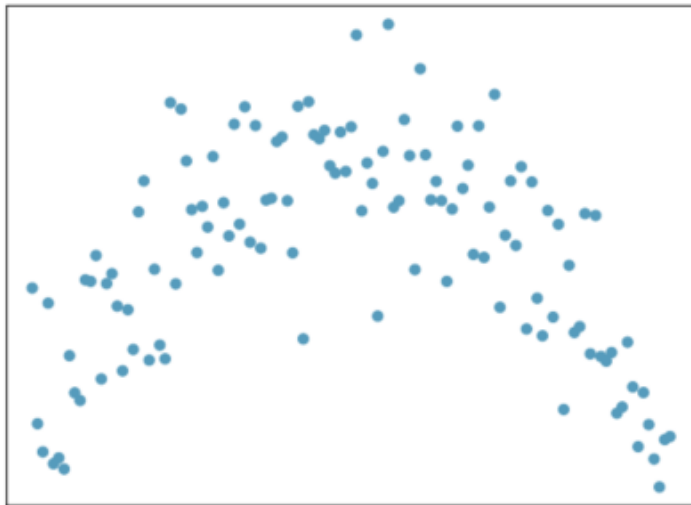
(a)



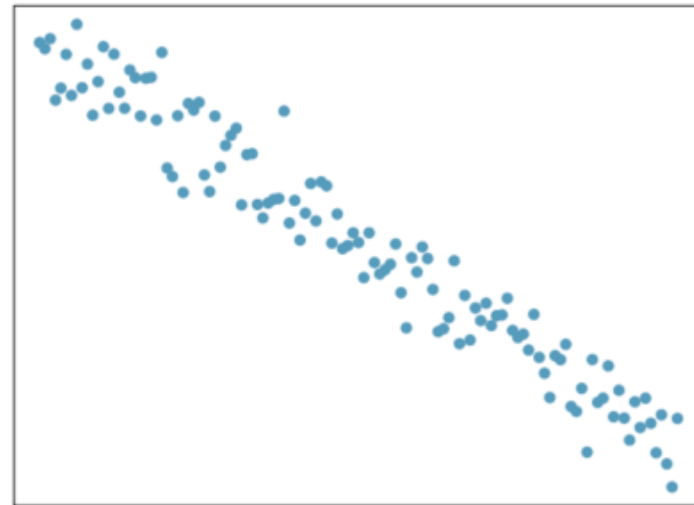
(b)



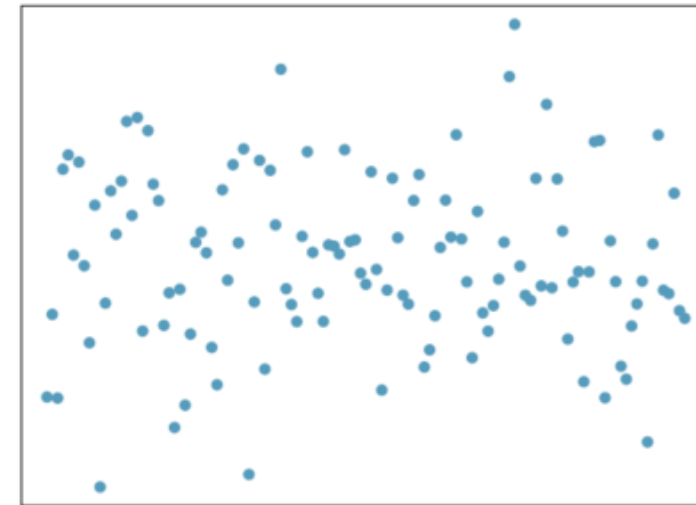
(c)



(d)

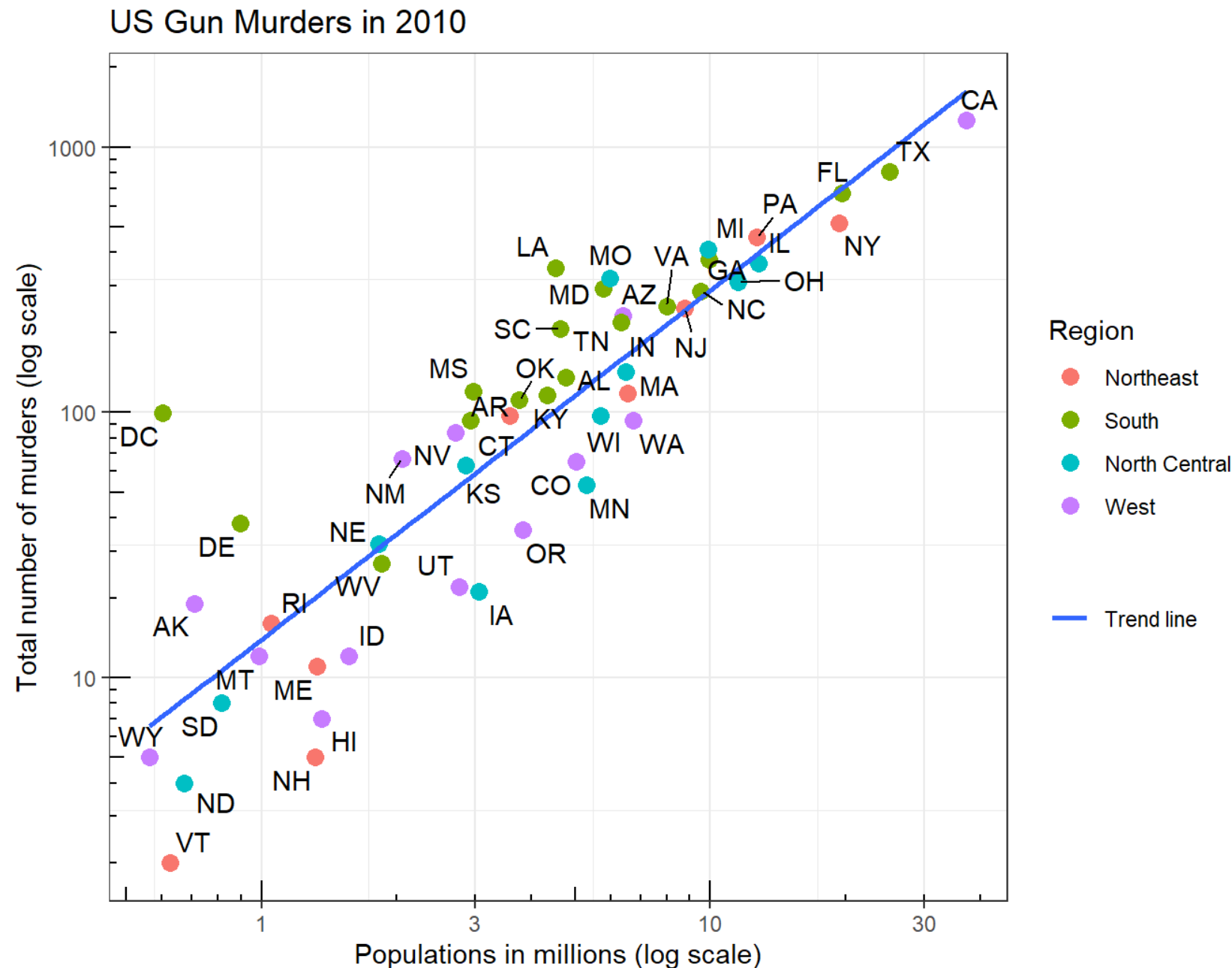


(e)



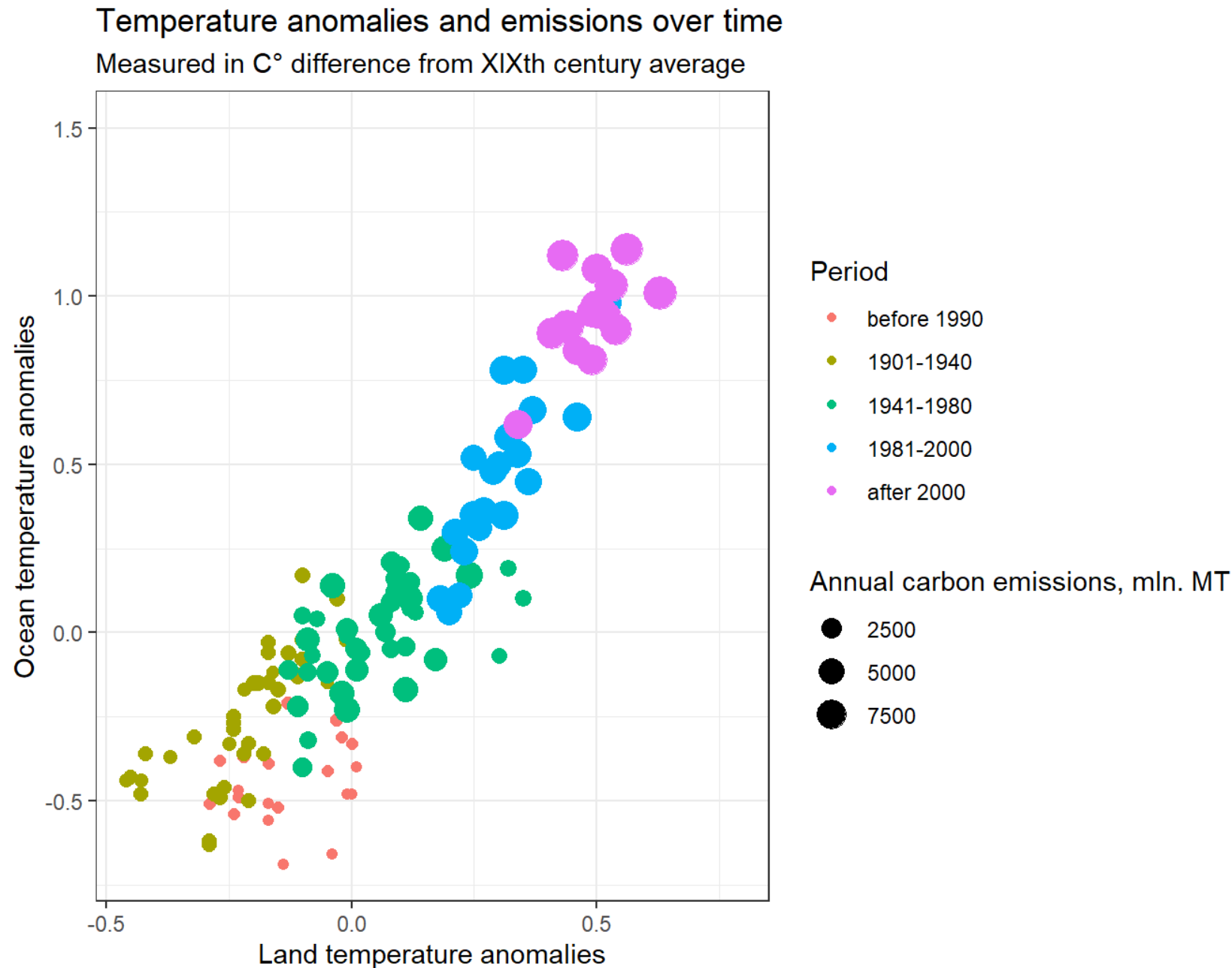
(f)

At home: More examples (1)



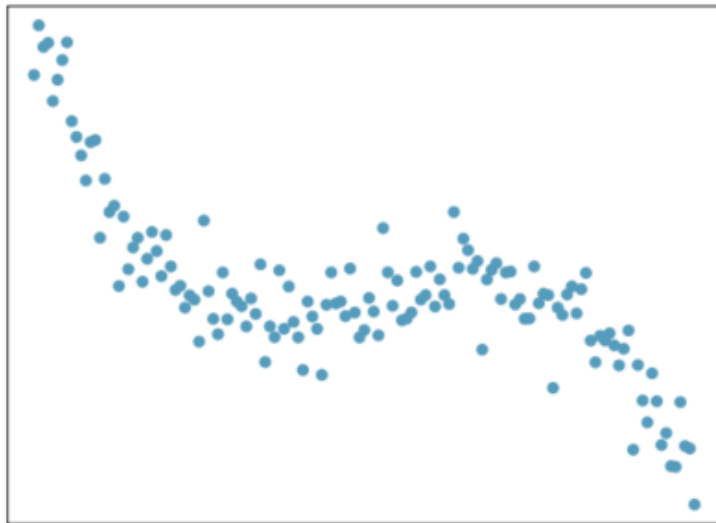
- Describe the scatter plot.
 - What is displayed. What is the plot type.
 - What is on X and Y axis. Are those transformed?
 - What points show. What shape/color stands for?
- Summarizes the relationship between Y and X.
 - Positive/negative.
 - Linear non-linear.

At home: More examples (2)

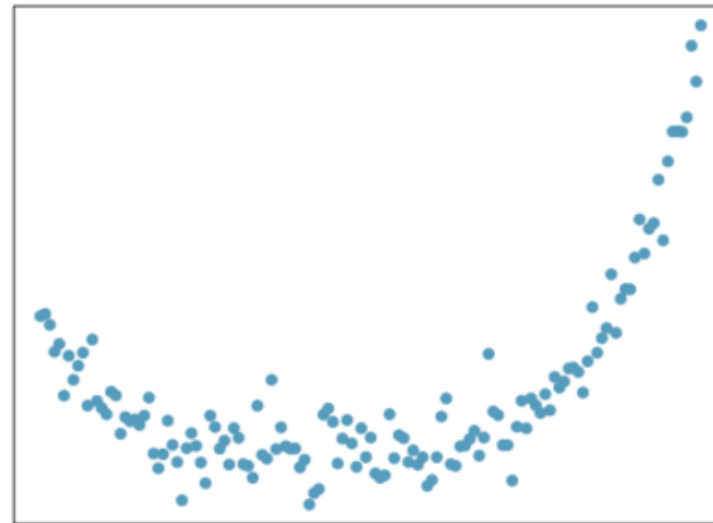


- Describe the scatter plot.
 - What is display. What is the plot type.
 - What is on X and Y axis. Are those transformed?
 - What points show. What shape/color stands for?
- Summarizes the relationship between Y and X.
 - Positive/negative.
 - Linear non-linear.

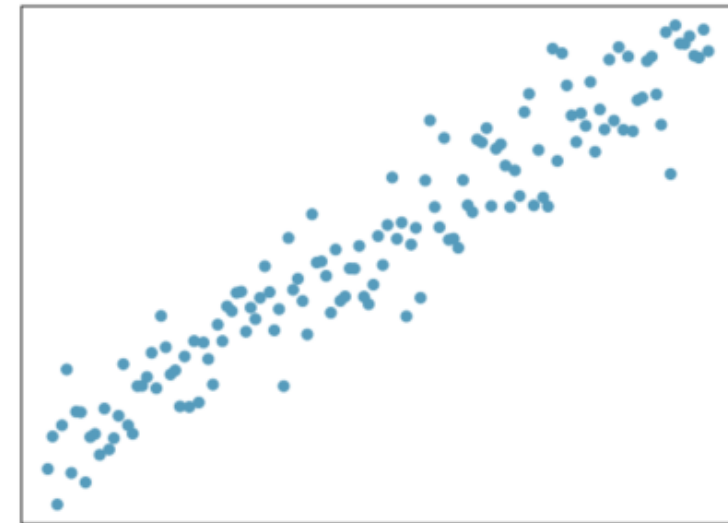
Example (exam): draw a linear trend line between x and y



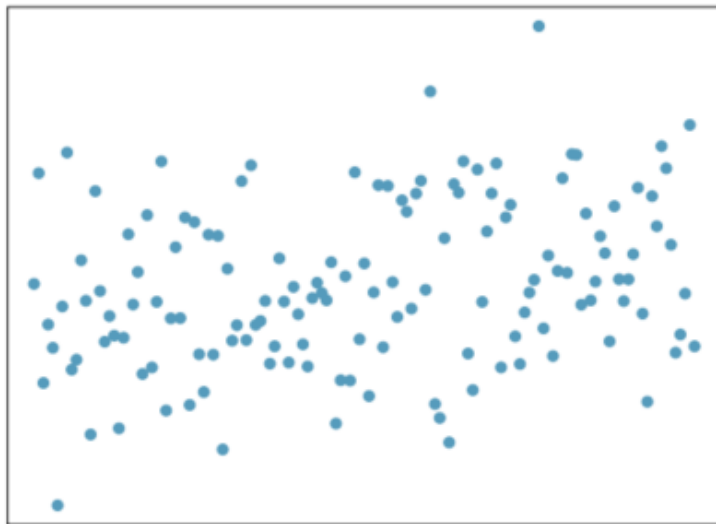
(a)



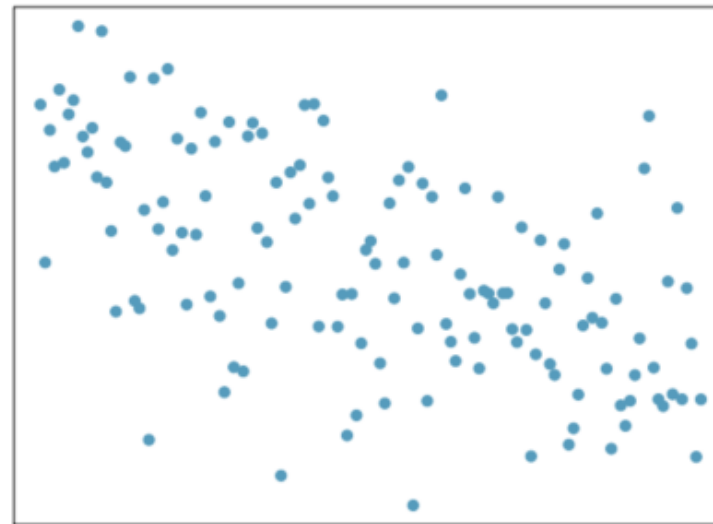
(b)



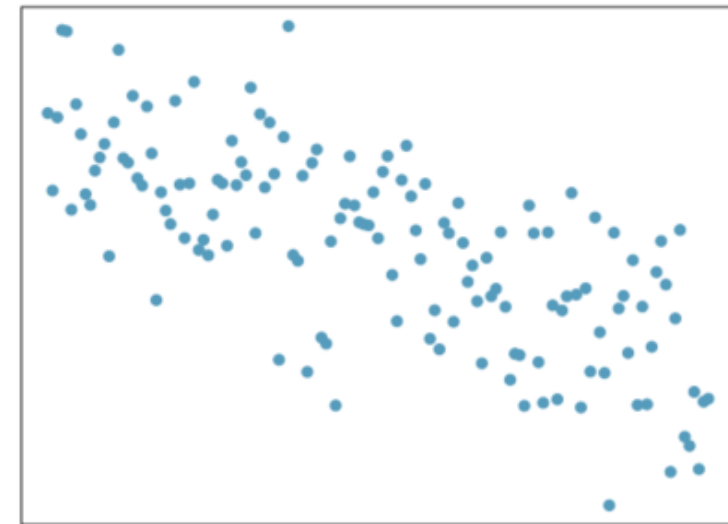
(c)



(d)



(e)



(f)

Correlation

is any statistical relationship, whether causal or not, between two random variables or bivariate data.

See more explanations:

- Josh Starmer: [Pearson's Correlation, Clearly Explained!](#)
- Khan Academy. [Unit: Exploring bivariate numerical data. Correlation coefficients](#)

Methodology

There are many [methods of correlation](#) estimation, for example:

Pearson's correlation coefficient

Population correlation

$$r_{xy} = \frac{1}{N} \sum_{i=1}^N \frac{x_i - \mu_x}{\sigma_x} \frac{y_i - \mu_y}{\sigma_y}$$

where:

- r_{xy} is the correlation coefficient;
- x_i and y_i are bi-variate observations;
- μ_x and μ_y are the population means;
- σ_x and σ_y are the population standard deviations;

Sample correlation

$$R_{xy} = \frac{1}{n-1} \sum_{i=1}^n \frac{x_i - \bar{x}}{s_x} \frac{y_i - \bar{y}}{s_y}$$

where:

- R_{xy} is the correlation coefficient;
- $\bar{x}$ and $\bar{y}$ are the sample means;
- s_x and s_y are the sample standard deviations;

Interpretation

Correlation coefficient captures **ONLY LINEAR** relationship.

Usually, it is interpreted arbitrary. [J. Cohen \(1988\)](#) defined it as:

- $r < 0.1$ Very small
- $0.1 \leq r < 0.3$ Small
- $0.3 \leq r < 0.5$
Moderate
- $r \geq 0.5$ Large

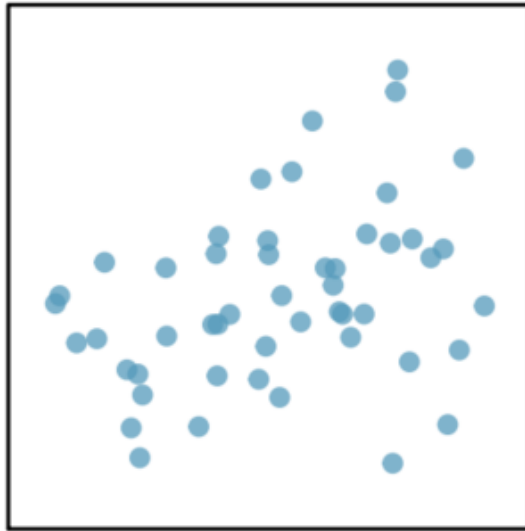
Positive correlation: $r > 0$

- x and y relate proportionally.
- An increase in x (y) is associated with an increase in y (x).

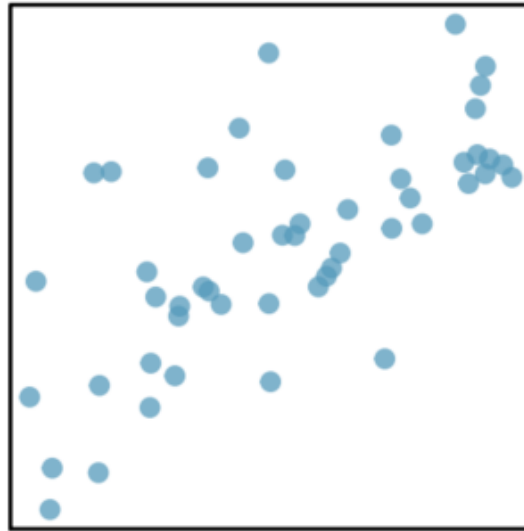
Negative correlation: $r < 0$

- x and y relate inversely;
- An **increase** in x (y) is associated with a **decrease** in y (x).

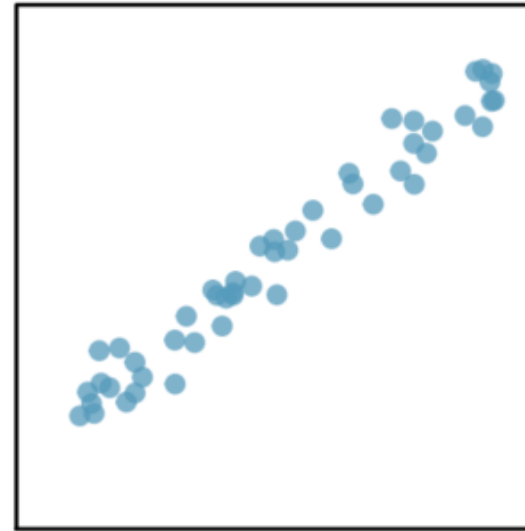
Correlation examples (1)



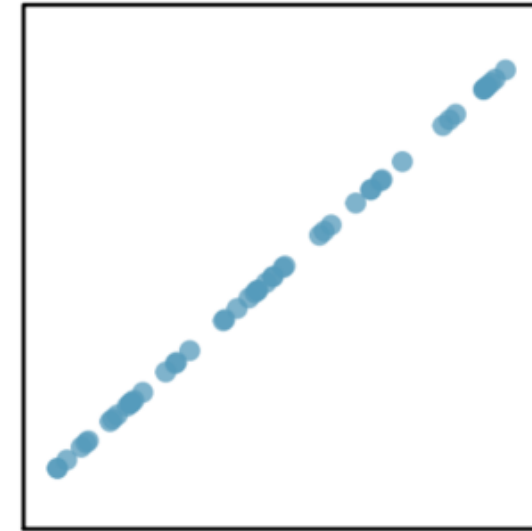
$R = 0.33$



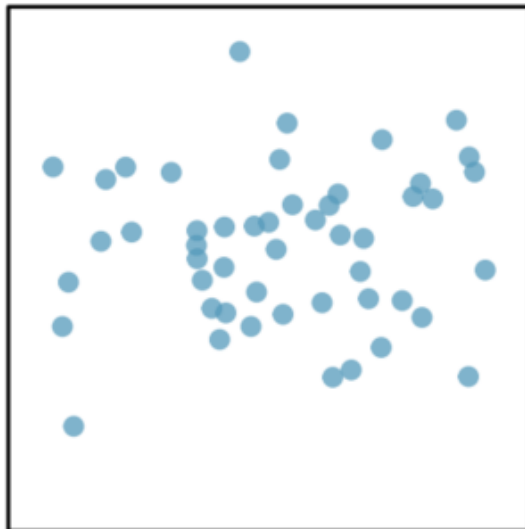
$R = 0.69$



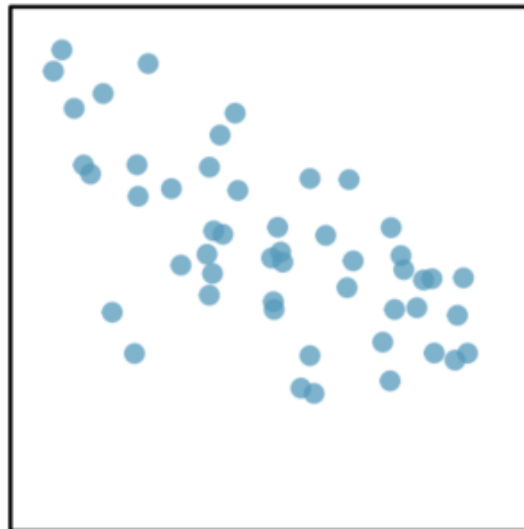
$R = 0.98$



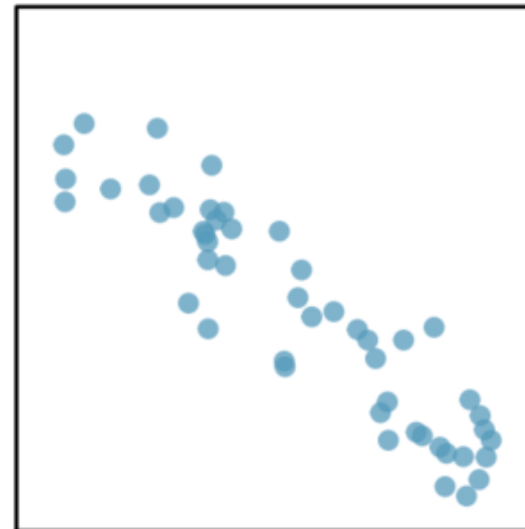
$R = 1.00$



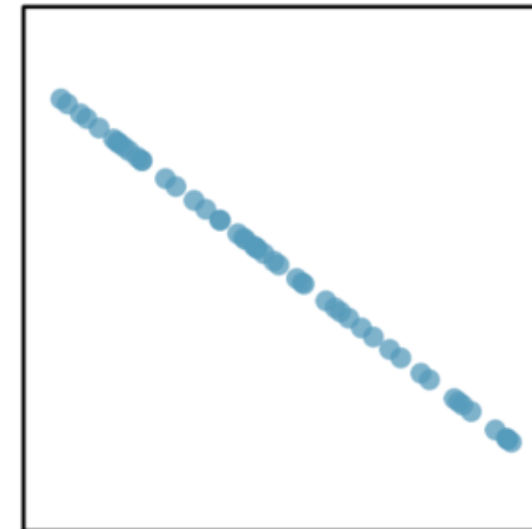
$R = 0.08$



$R = -0.64$

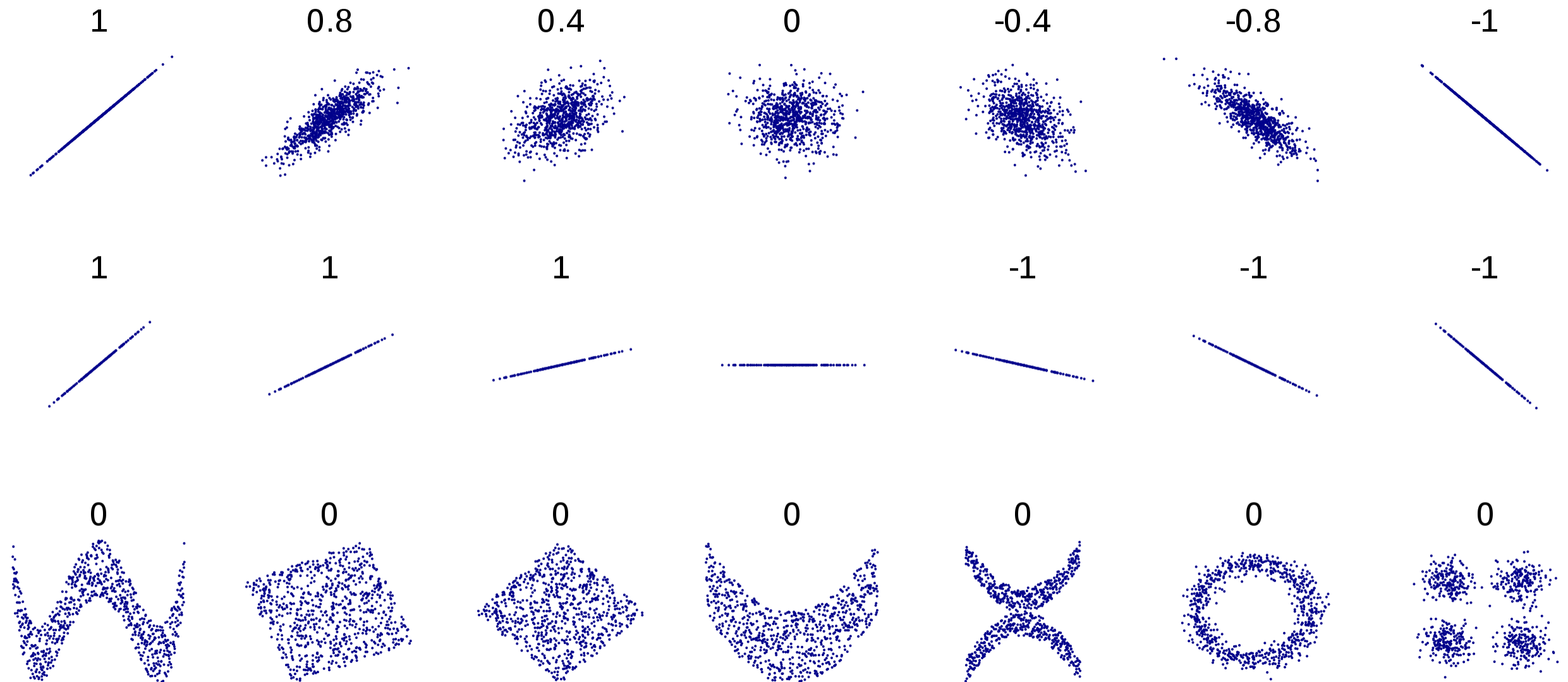


$R = -0.92$



$R = -1.00$

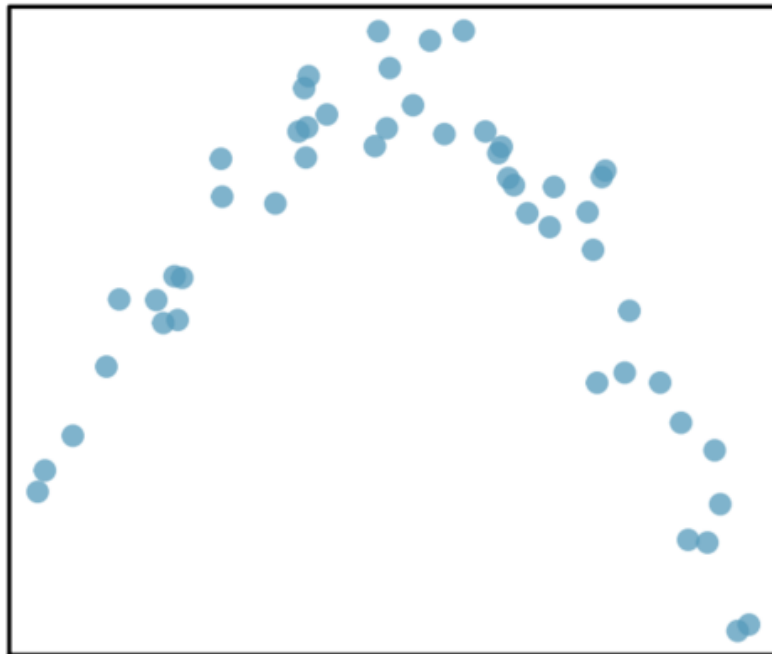
Correlation examples (2)



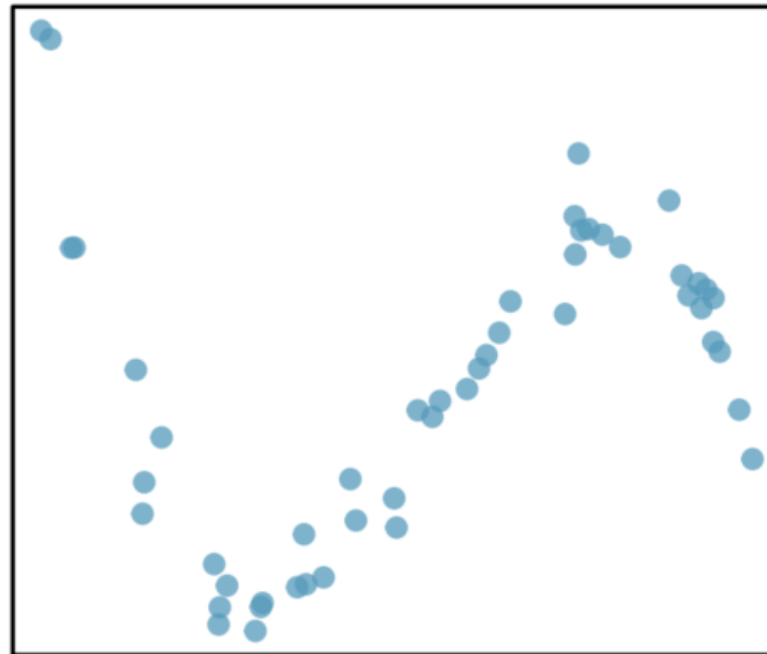
Correlation pitfalls: Non linearity

Correlation cannot be used with non-linear data.

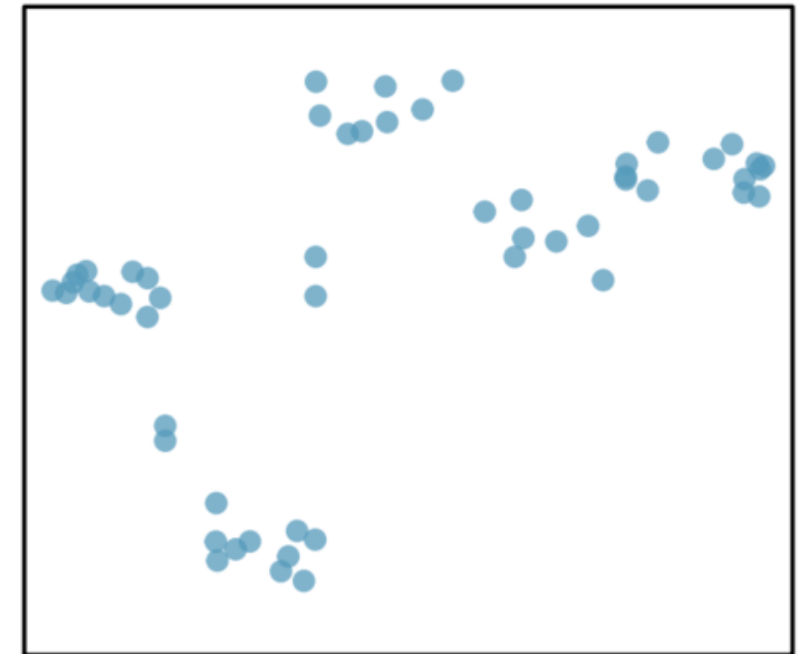
- Non linear data yields a correlation coefficient, but such coefficient does not imply any linear relationship.



$R = -0.23$

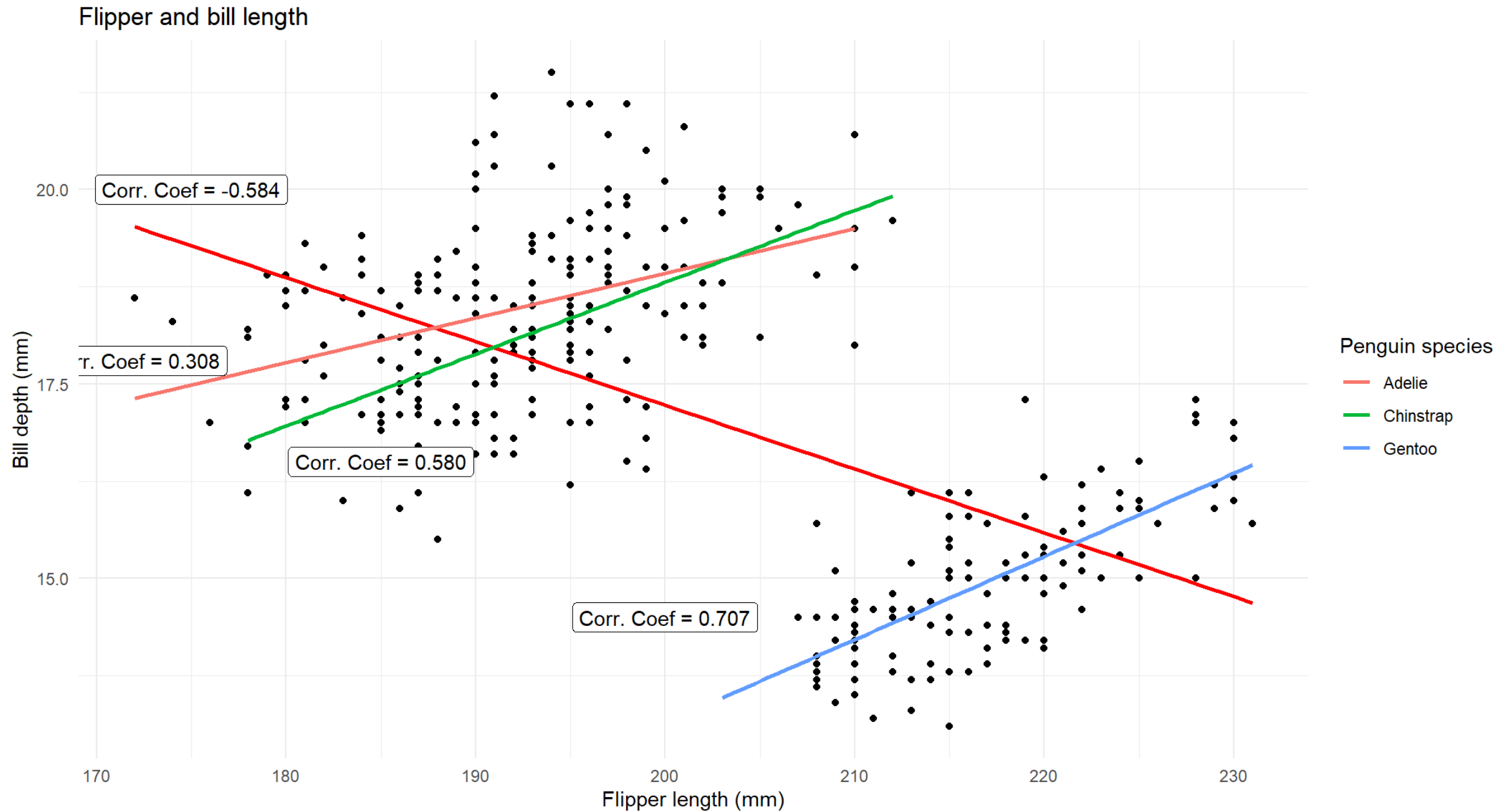


$R = 0.31$



$R = 0.50$

Correlation pitfalls



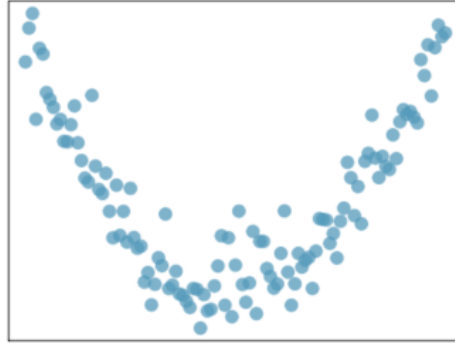
Correlation matrix (1): Penguins

	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g
bill_length_mm	1.000	-0.235	0.656	0.595
bill_depth_mm	-0.235	1.000	-0.584	-0.472
flipper_length_mm	0.656	-0.584	1.000	0.871
body_mass_g	0.595	-0.472	0.871	1.000

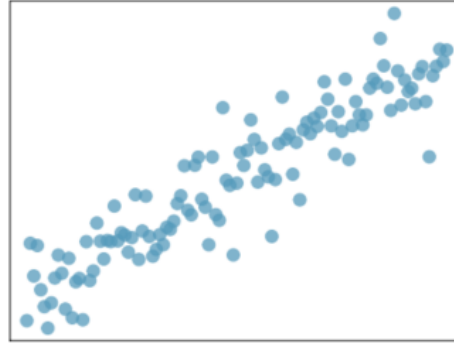
Examples (exam) on correlation

Match a correlation coefficient and a plot

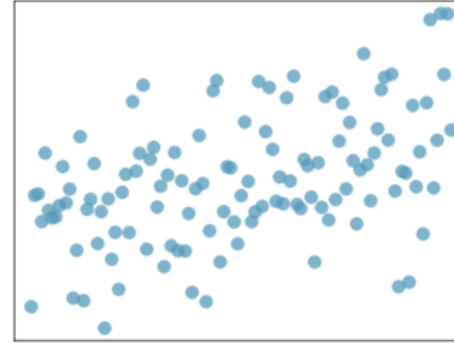
- (a) $R = -0.7$
- (b) $R = 0.45$
- (c) $R = 0.06$
- (d) $R = 0.92$



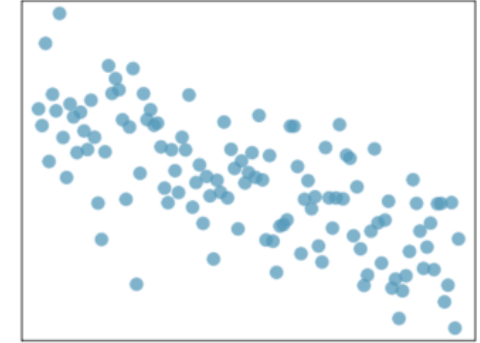
(1)



(2)



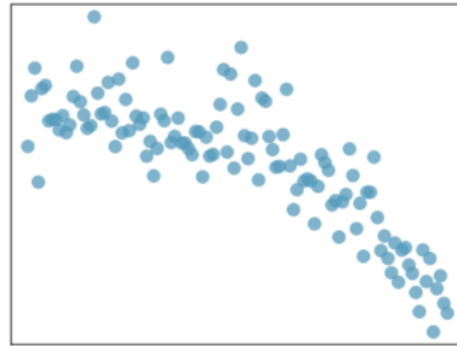
(3)



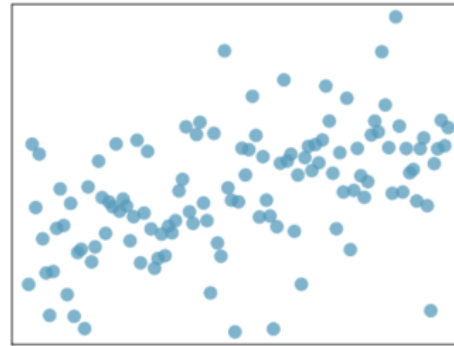
(4)

At home: Match a correlation coefficient and a plot

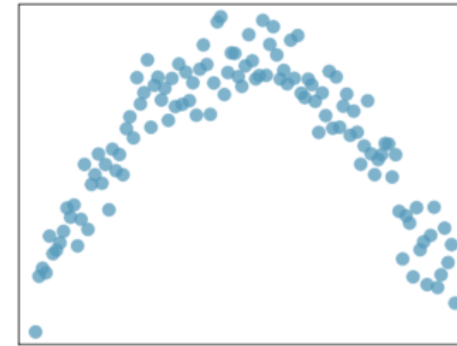
- (a) $R = 0.49$
 (b) $R = -0.48$
 (c) $R = -0.03$
 (d) $R = -0.85$



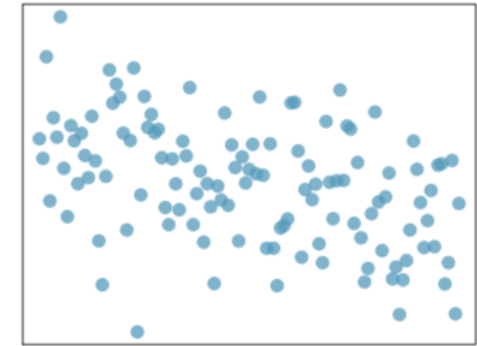
(1)



(2)

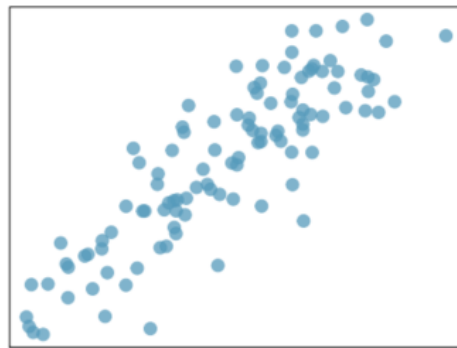


(3)

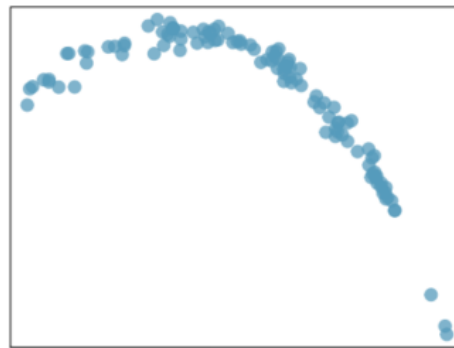


(4)

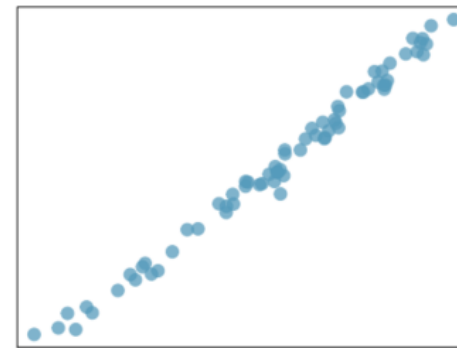
- (a) $r = -0.72$
 (b) $r = 0.07$
 (c) $r = 0.86$
 (d) $r = 0.99$



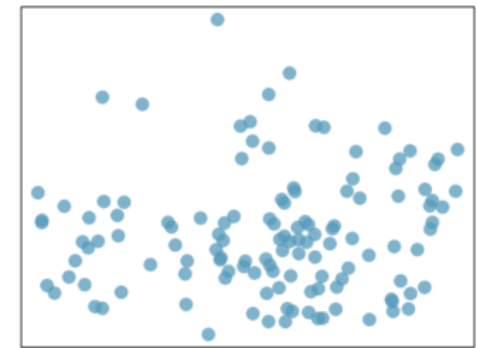
(1)



(2)

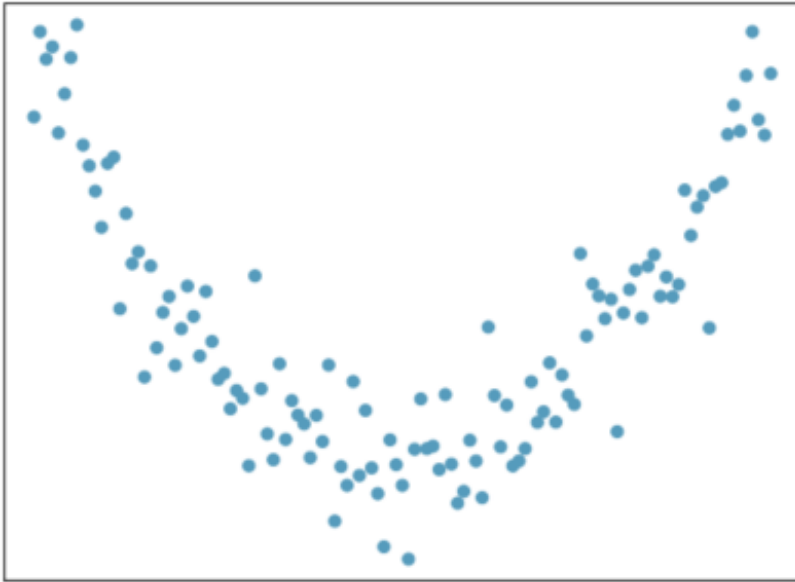


(3)

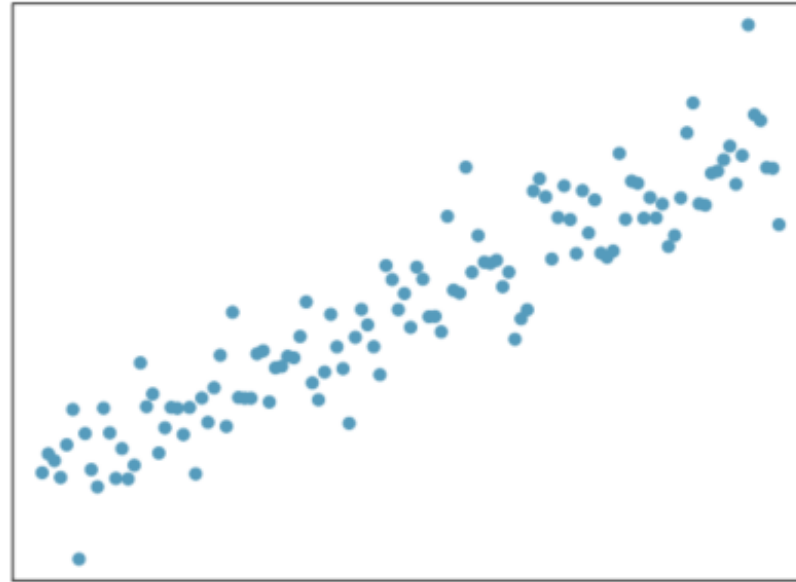


(4)

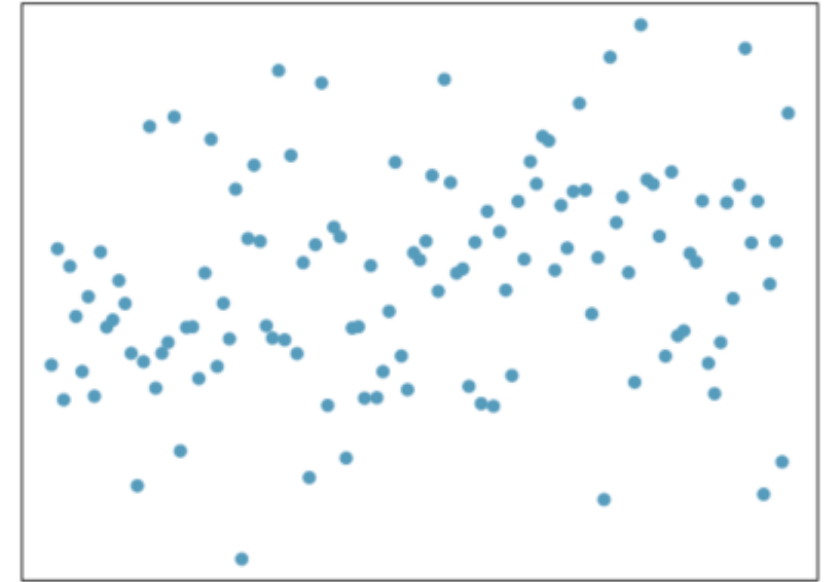
At home: Identify the relationship and its strength (1)



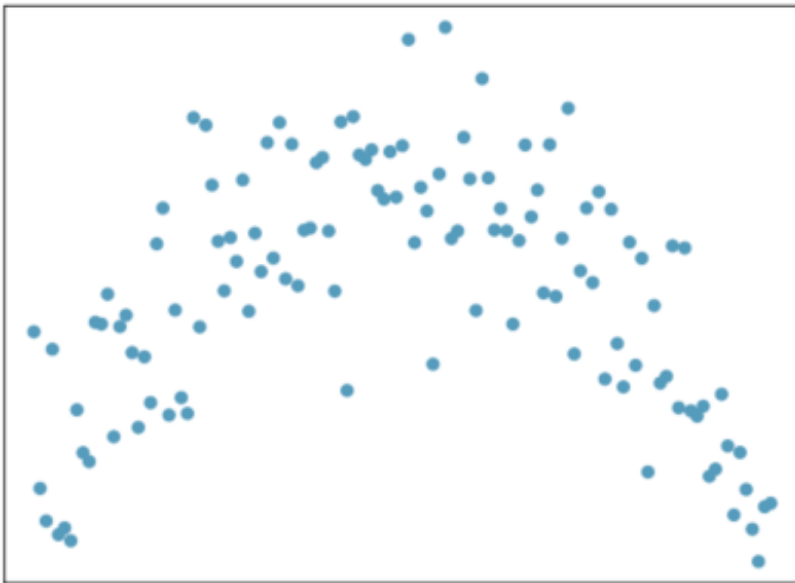
(a)



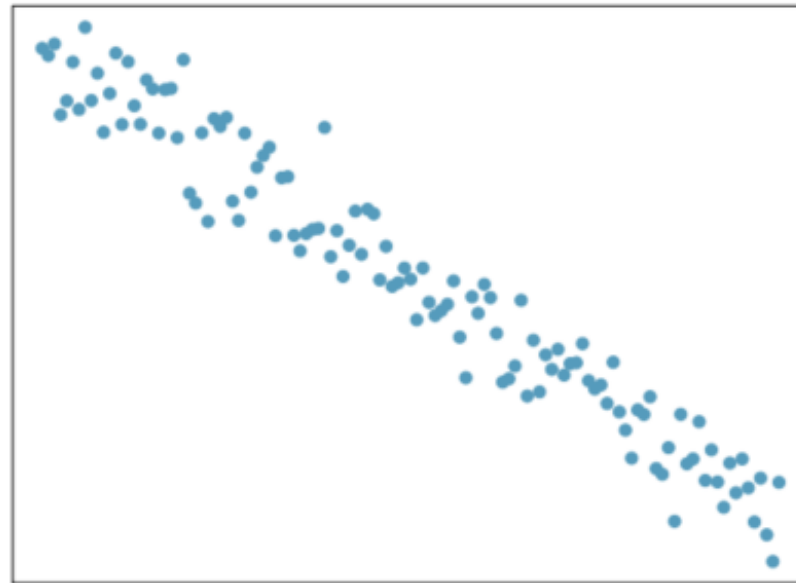
(b)



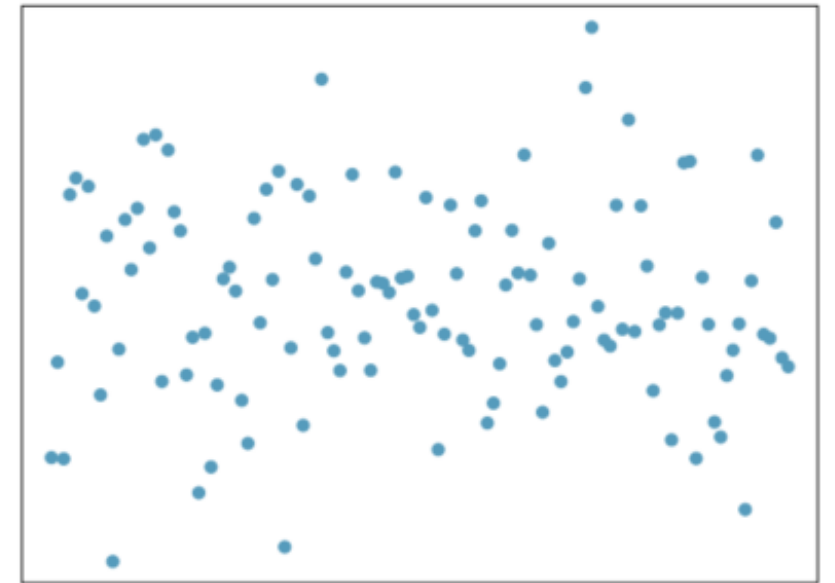
(c)



(d)

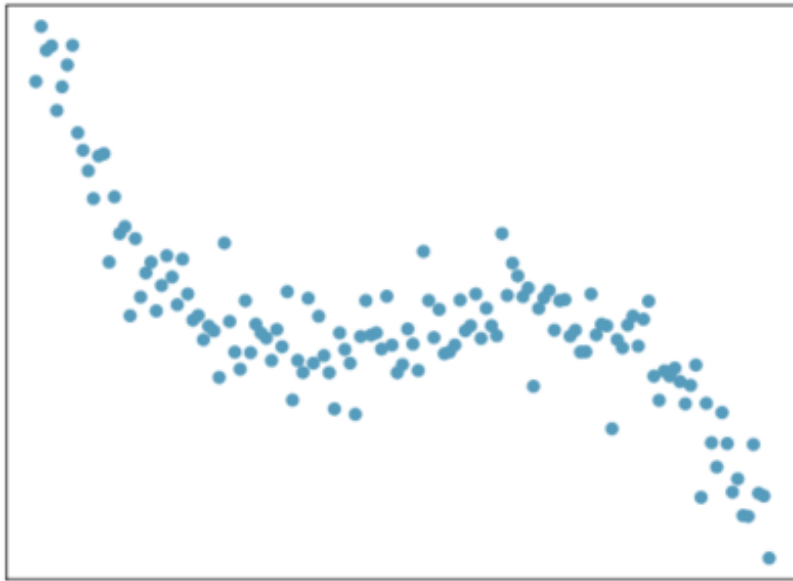


(e)

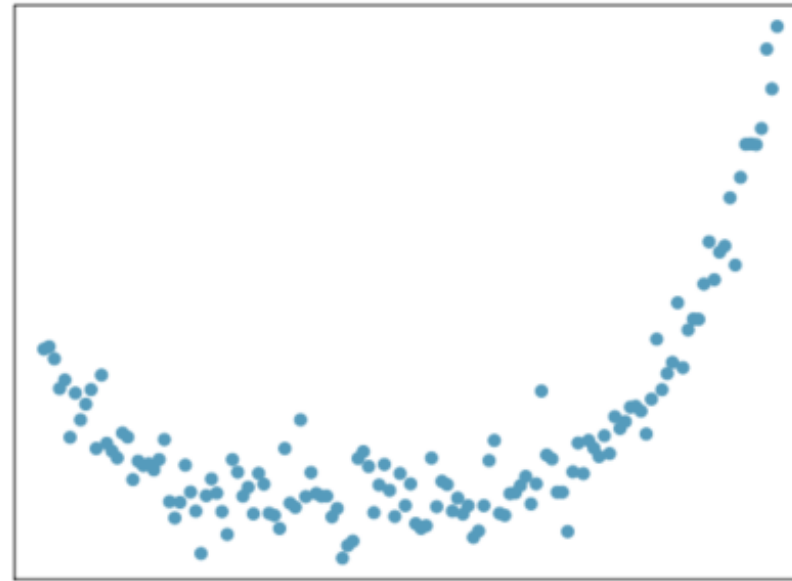


(f)

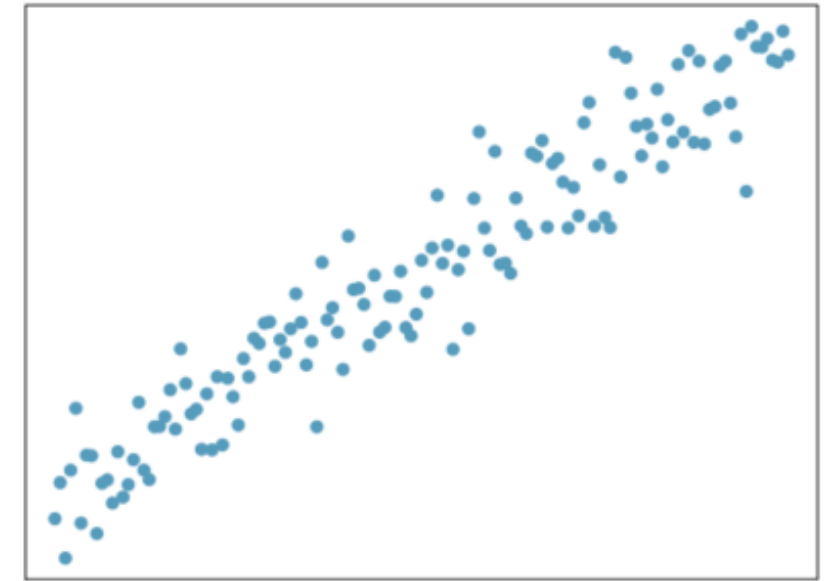
At home: Identify the relationship and its strength (2)



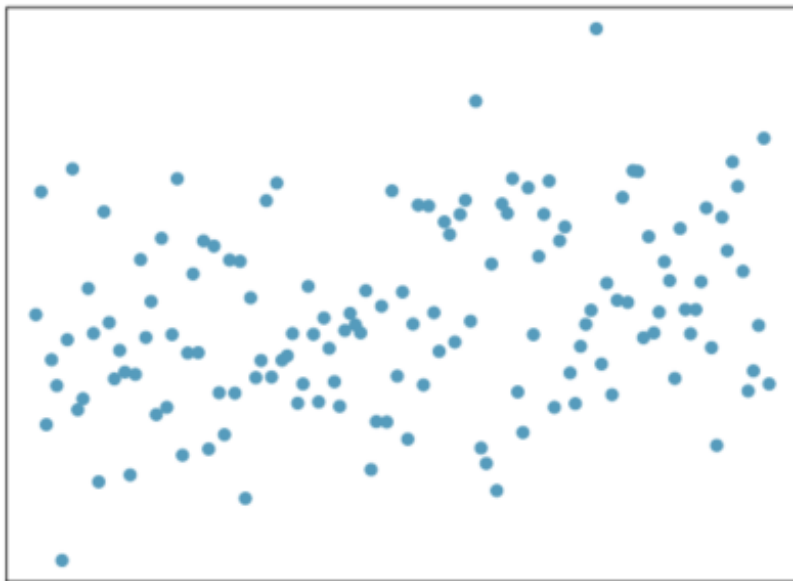
(a)



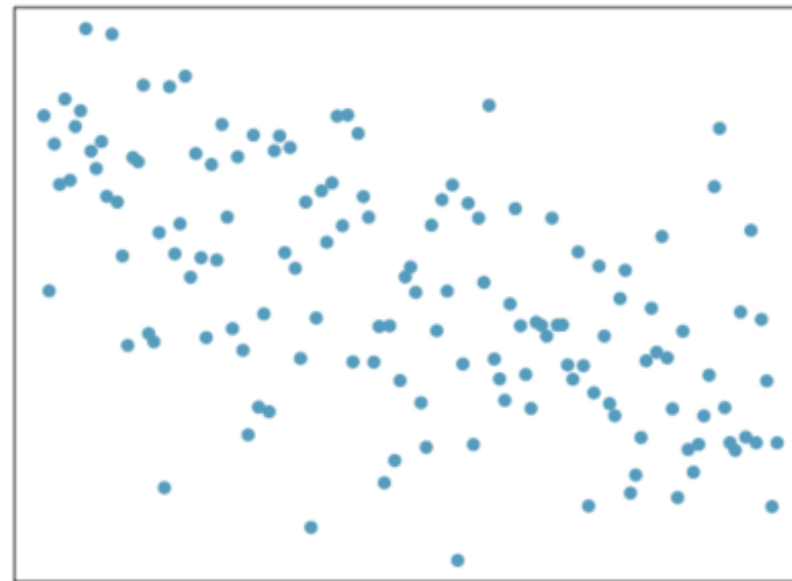
(b)



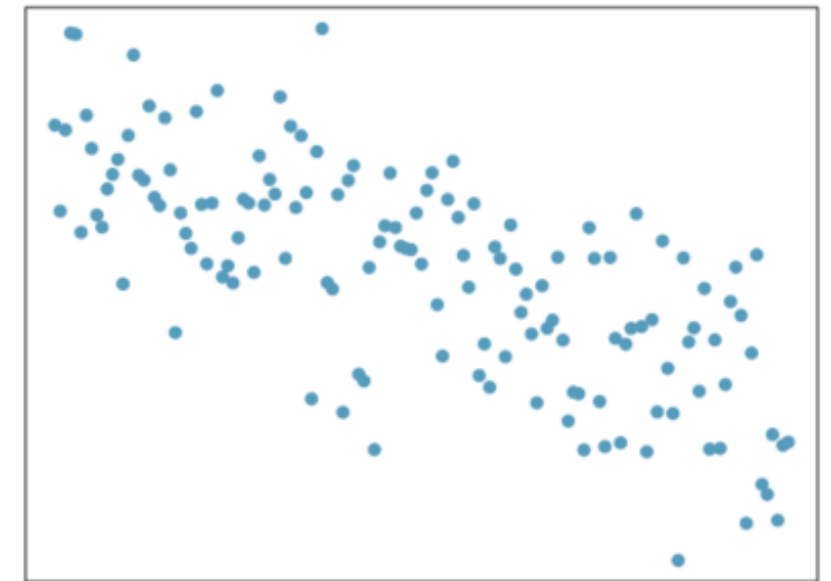
(c)



(d)



(e)



(f)

Spurious correlation (link)

Explore at home on your own:

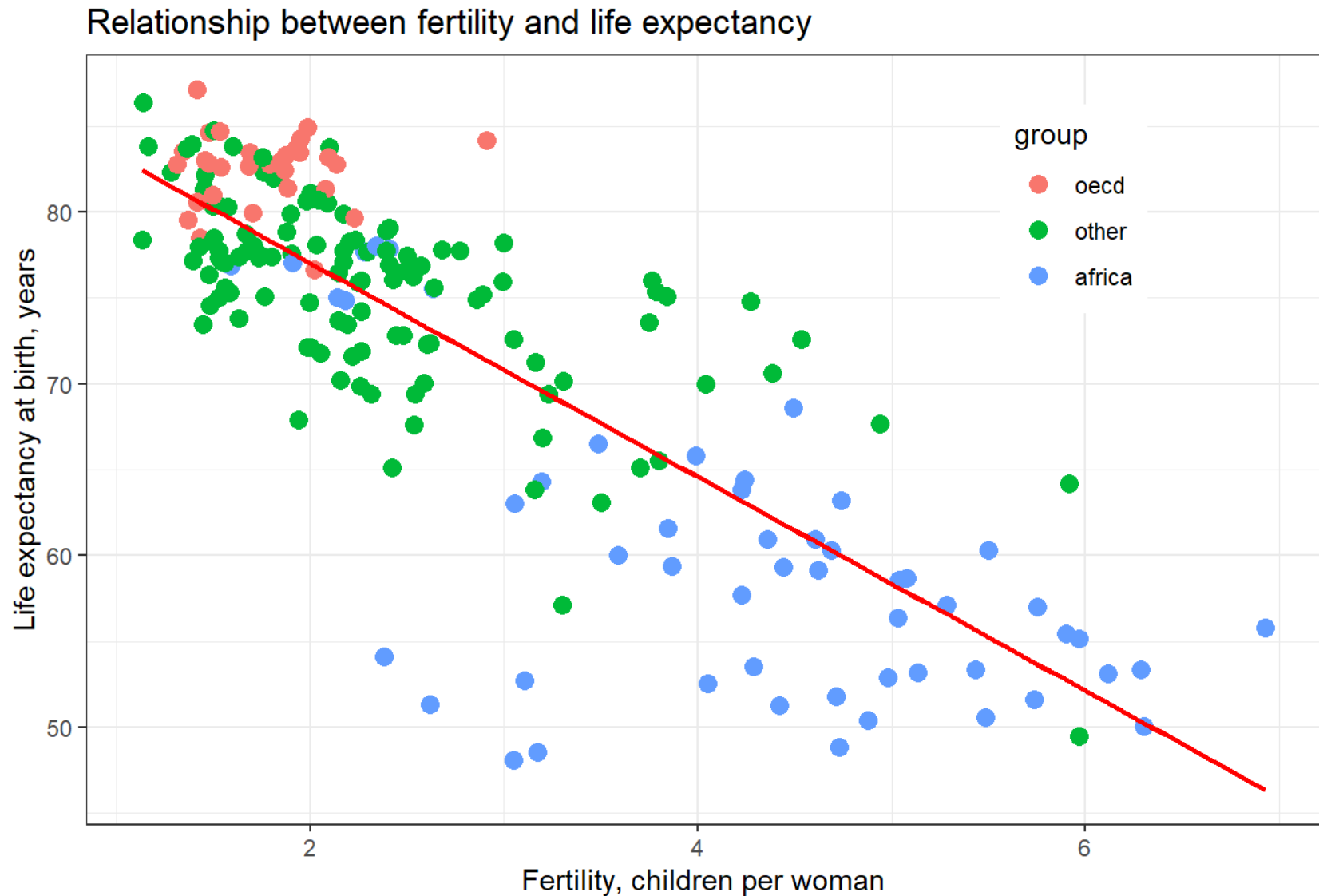
- <https://www.tylervigen.com/spurious-correlations>
- Define what is “spurious correlation”?

Correlation and Causation

If x correlates with y , does it mean that x causes y ?

- Correlation does not imply causation!

Do we live longer because of the birth rate?



Data source: Weisberg, S. (2014). Applied Linear Regression

The correlation between Fertility and life expectancy is -0.824.

- What does this mean?
- Can we imply that giving fewer births will increase life expectancy?
- What does the linear trend suggests?
- Does being from a different country groups causes this trend?

No... Non of it implies causality.

To establish a causal relationship

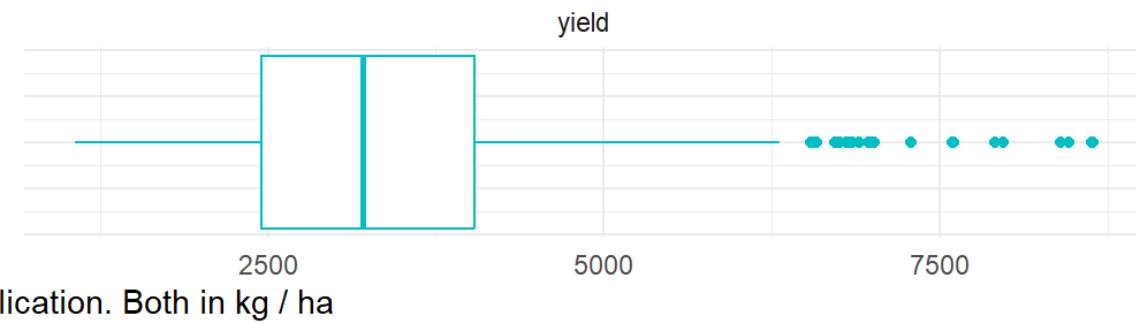
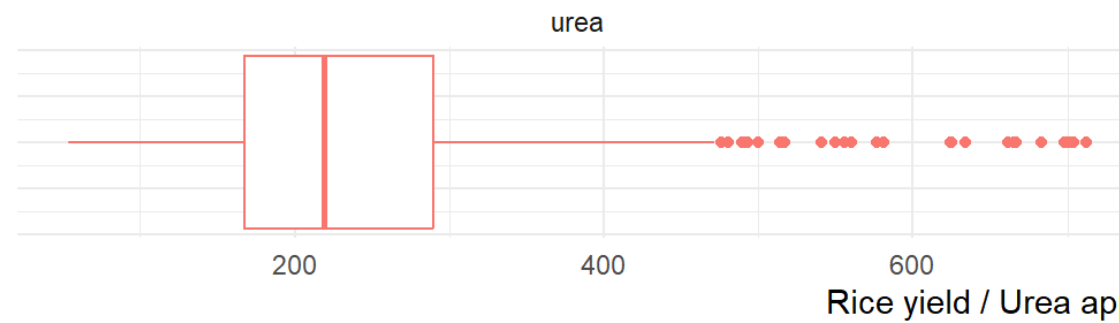
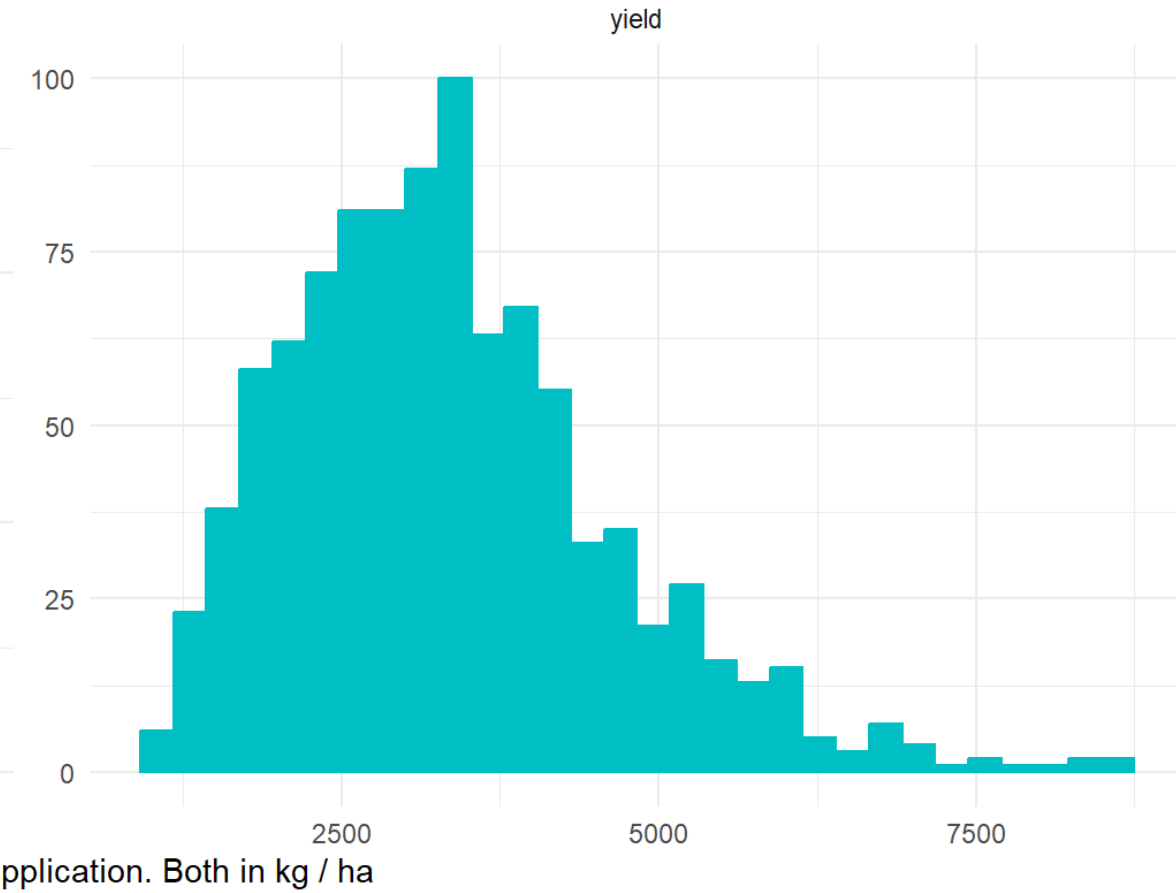
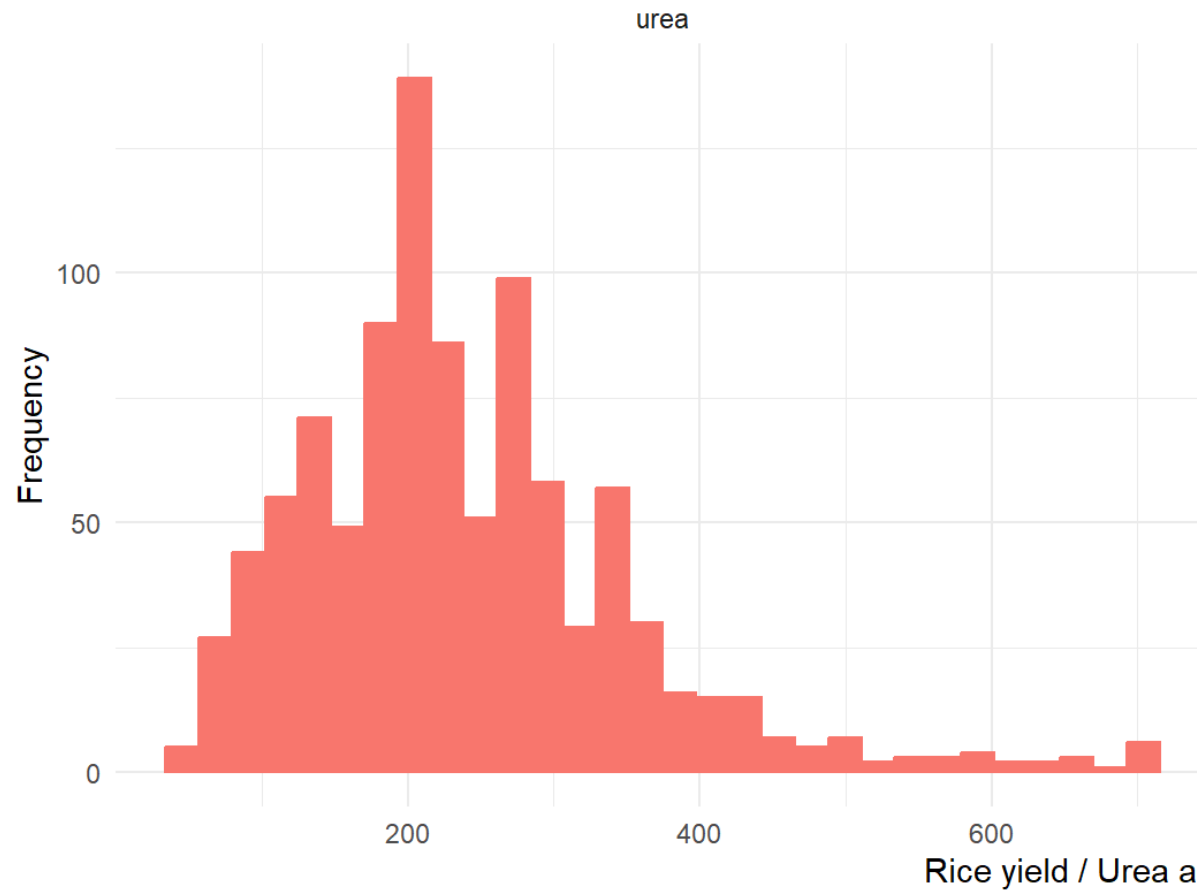
- We need to create a “Ceteris Paribus”!
- We need to make sure that “Other things” are equal between the countries.
- For example:
 - All countries consist of identical people, with identical sociology-economic characteristics.
 - The only difference is in their fertility rate and (perhaps) life expectancy.

Exploratory analysis example: rice yields and fertilizers application

How does the urea application affect the yields of rice?

- Sample of 171 Rice farms in Indonesia is followed over 5 years.
- Each observation represents an individual farm that produced rice over one year.
- Two variables:
 - Rice yields in kg / ha
 - Urea ($CO(NH_2)_2$) application in kg / ha

Univariate analysis



Descriptive Statistics

Descriptive statistics for the full sample

Variable	Mean	SD	Median	Min	Max
yield	3381.8	1676.3	3160.0	400.0	27500
urea	236.9	140.1	214.3	0.9	2000

- What about the influential observations/outliers?

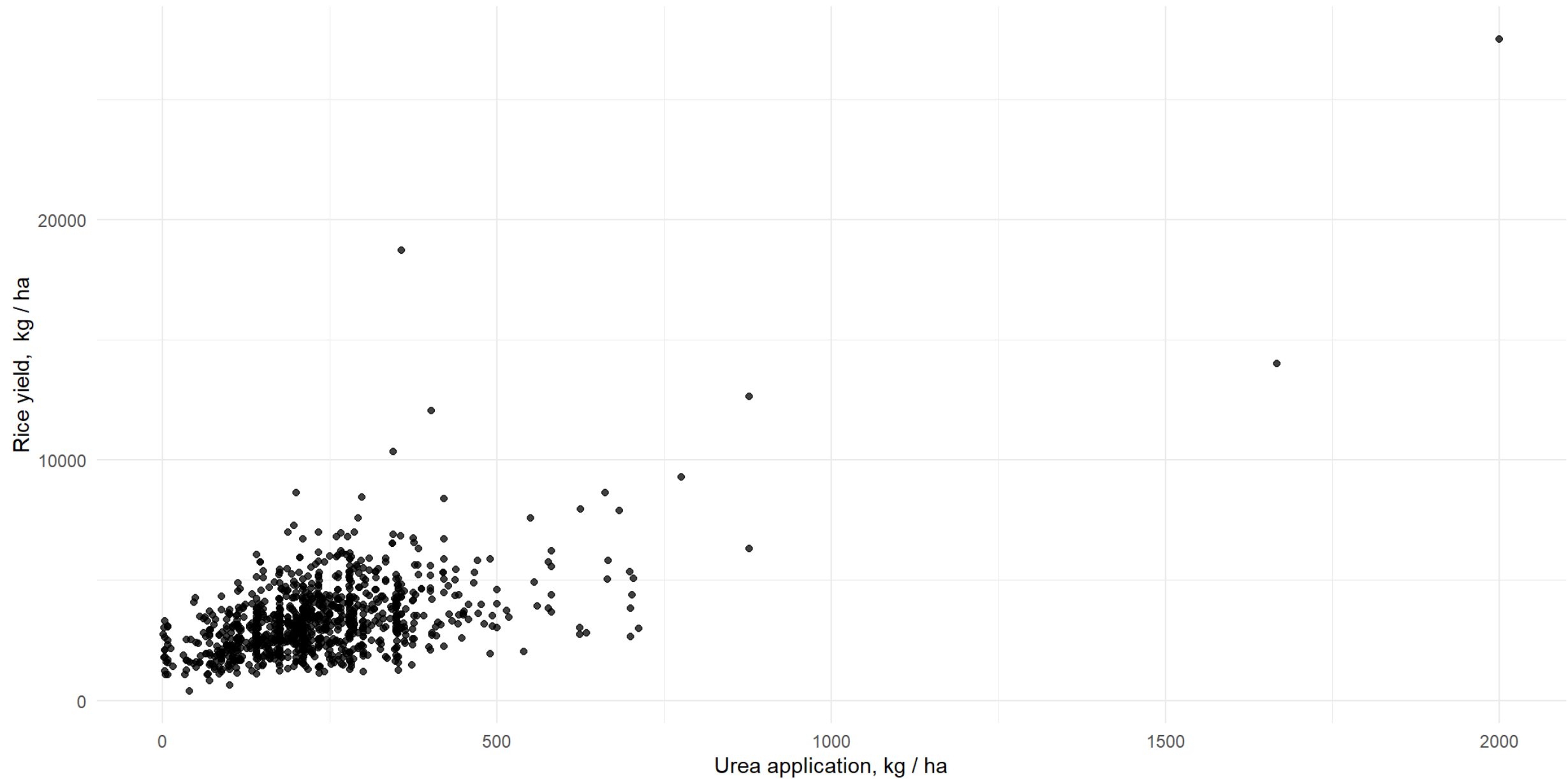
Descriptive statistics after excluding suspected outliers

Variable	Mean	SD	Median	Min	Max
yield	3348.8	1276.7	3207.0	1066.7	8640.0
urea	239.4	111.2	218.9	53.8	712.3

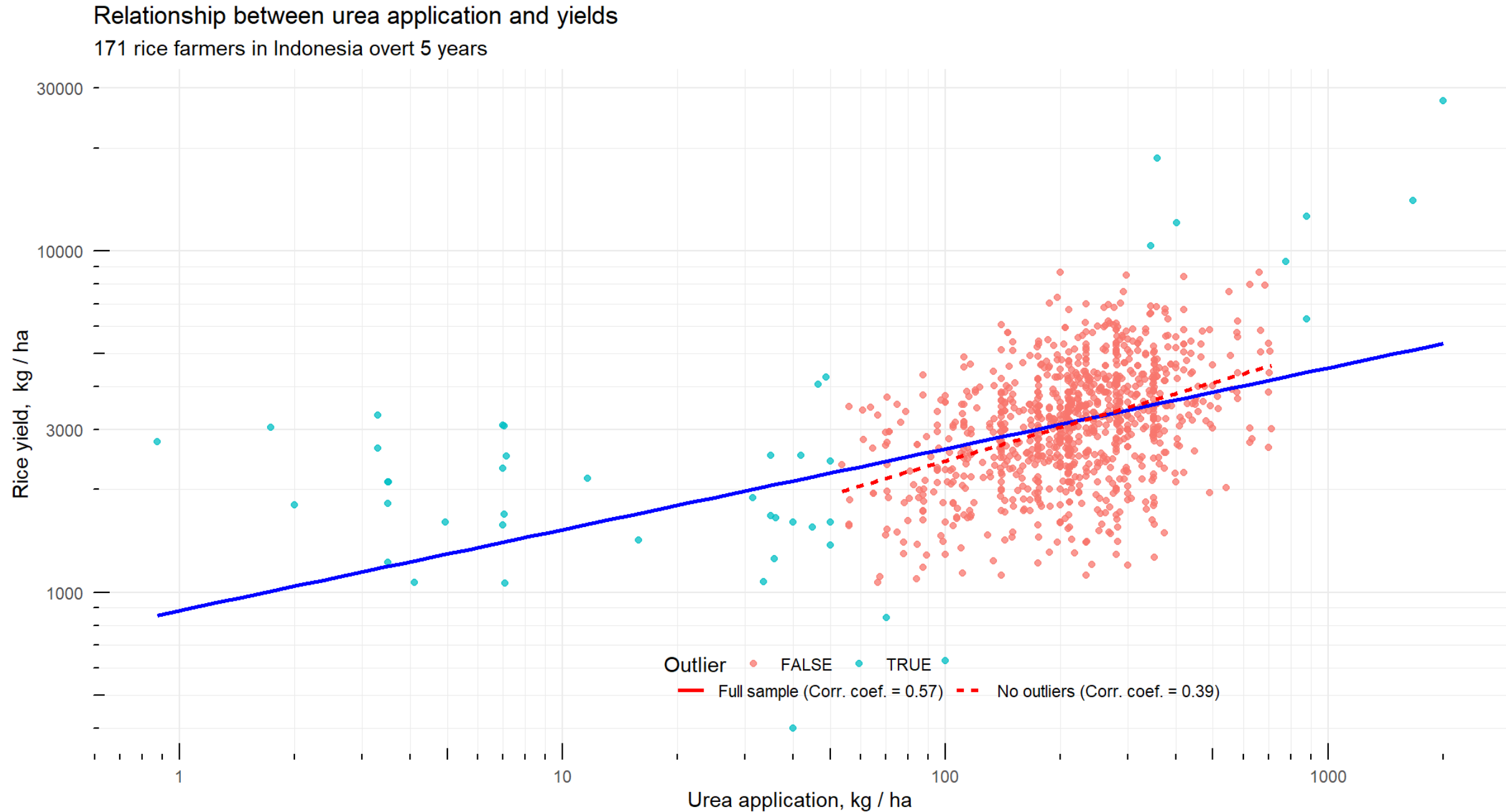
Scatter plot

Relationship between urea application and yields

171 rice farmers in Indonesia over 5 years



Linearity and linear transformation



Correlation (exam questions examples)

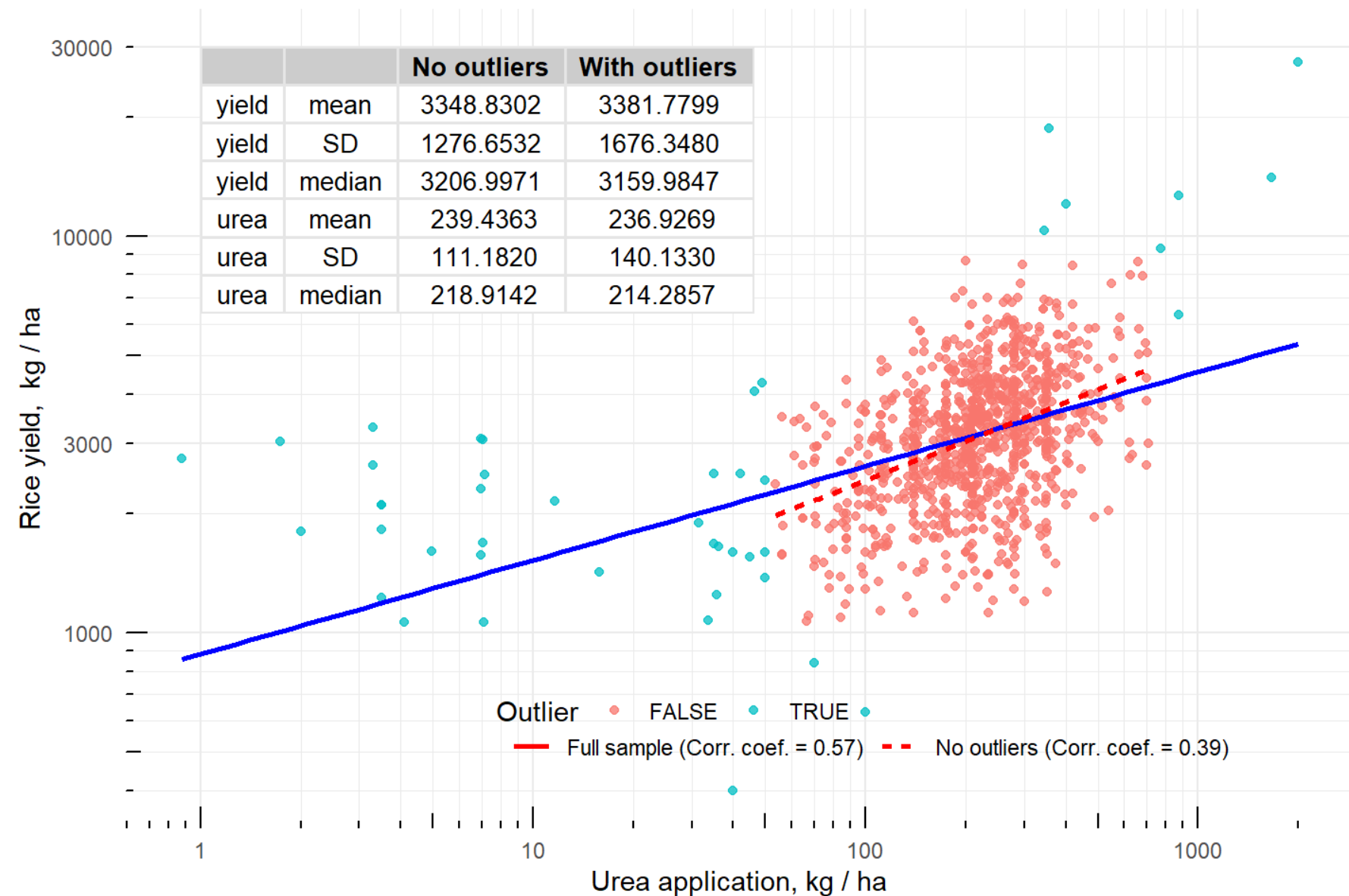
What can we conclude about the relationship?

- Linear / non-linear; strong / weak.
- What is the central tendency and variance?
- What is the effect of outliers on the central tendency and variance?

Does the correlation suggests that urea increases the rice yield in Indonesia?

- Explain why!

Relationship between urea application and yields
171 rice farmers in Indonesia overt 5 years



Causality conclusion

Does the correlation suggest that urea application increases the yield of rice in Indonesia?

- In theory, fertilizers should increase yields. But:
 - Increase in N without other fertilizers (P and K) may not increase yield because of the [Liebig's law of the minimum](#).
 - Water availability, technology and other production factors may affect the yields.
 - Climate: El Niño and La Niña
 - Soil and seeds, ...
- As data contains different farmers over different years, these factors are likely to be different!

Conclusion:

In the current setting, we **CANNOT** conclude that an increase in application of urea **causes** an increase in rice yields because:

- **other factors are not constant** between different farms/observations/periods. See those factors above.
- or **there is no Ceteris Paribus**

Cause, Effect, Potential Outcome and Ceteris Paribus

The econometrician's goal is to :

- infer that one variable (such as fertilizers)
- has a causal effect on another variable (such as yields).

Association between two or more variables might be suggestive, but not causal ([Wooldridge 2020](#)).

Cause and Effect in Rice Yields

1. The Cause or the **Treatment**

- in 2014, farm A uses 100 kg of urea instead of 0: $\text{Urea}_A = 100$

2. Outcome or **Factual**

- rice yields in 2015
- $\text{Yield}_A(\text{Rice}|\text{Urea} = 100)$

3. **Effect**

- How to measure it?

$$\text{Effect} = \text{Outcome} - \text{Potential Outcome}$$

2.a Potential Outcome, or **Counterfactual**

- in 2014, farm A **did not** increase urea use;
- potential rice yields in 2015
- $\text{Yield}_A(\text{Rice}|\text{Urea} = 0)$

$$\text{Effect} = \text{Factual} - \text{Counterfactual}$$

$$\text{Causal effect} = \text{Yield}_A(\text{Rice}|\text{Urea} = 100) - \text{Yield}_A(\text{Rice}|\text{Urea} = 0)$$



Once the farm A decide to apply Urea, can we possible learn the Potential Outcome?

What we usually measure:

Farm A

- $\text{Urea}_A = 100$
- $\text{Yield}_A(\text{Rice} | \text{Urea} = 100)$

Farm B

- $\text{Urea}_B = 0$
- $\text{Yield}_B(\text{Rice} | \text{Urea} = 0)$

We usually observe just a difference between two factuais:

$$\text{Yield difference} = \text{Yield}_A(\text{Rice} | \text{Urea} = 100) - \text{Yield}_B(\text{Rice} | \text{Urea} = 0)$$

This is not the same as the causal effect because **the other things are not equal** between farms A and B.

Cause or Treatment

- Variable or input that we change and which effect we want to understand.

Outcome

- Measured characteristics that result from a treatment.

Causal Effect

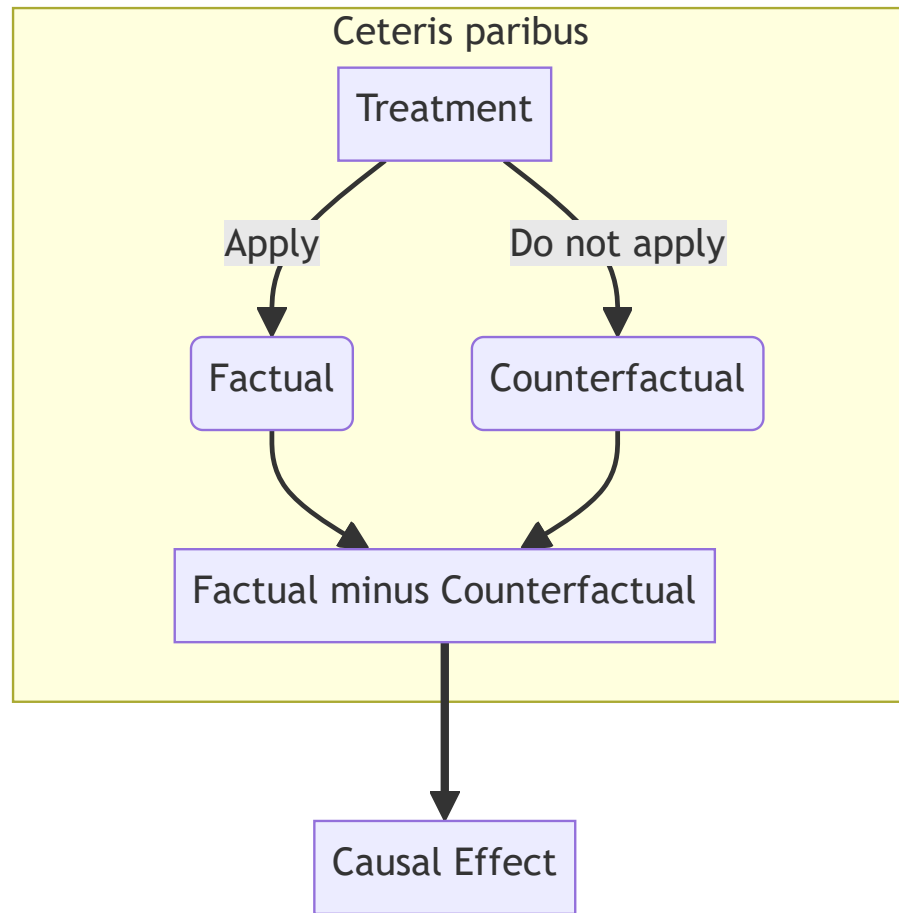
- Difference between **factual and counterfactual** (potential outcome).

Potential Outcome or Counterfactual

An outcome that **could have been observed** if we had not applied the treatment.

- we only observe factual outcomes.
- we cannot observe counterfactual
 - because once the treatment is applied, subject cannot be observed without the treatment.
- counterfactual is the potential outcome of the treatment, where all other things except for the treatment are equal / **given ceteris paribus**.

Cause, Effect and the Ceteris Paribus



To reveal the effect caused by a treatment, we need to ensure the ceteris paribus.

The “other things” are:

- **measurable features** (age, height, costs, ...);
- **factors that are measured but not included in our data;**
- **unmeasurable factors** (ability, effort, connections, ...)

How to establish the ceteris paribus?

There are 5 key statistical tools that help us doing so.

1. **RCT - Randomized Control Trials**
2. **Regression**
3. IV - Instrumental Variable
4. DID - Difference in Difference (Panel Regression)
5. RDD - Regression Discontinuity Design

At home: New insights on the HIV pandemic

The purpose of the following short video is to stimulate debate on what are **covariates, and causes** of the HIV pandemic in the world.

After watching it, discuss:

- Facts of the HIV/AIDS pandemic;
- Country-specific peculiarities that correlate with the pandemic.
- What causes the pandemic to spread at so different rates across the countries?

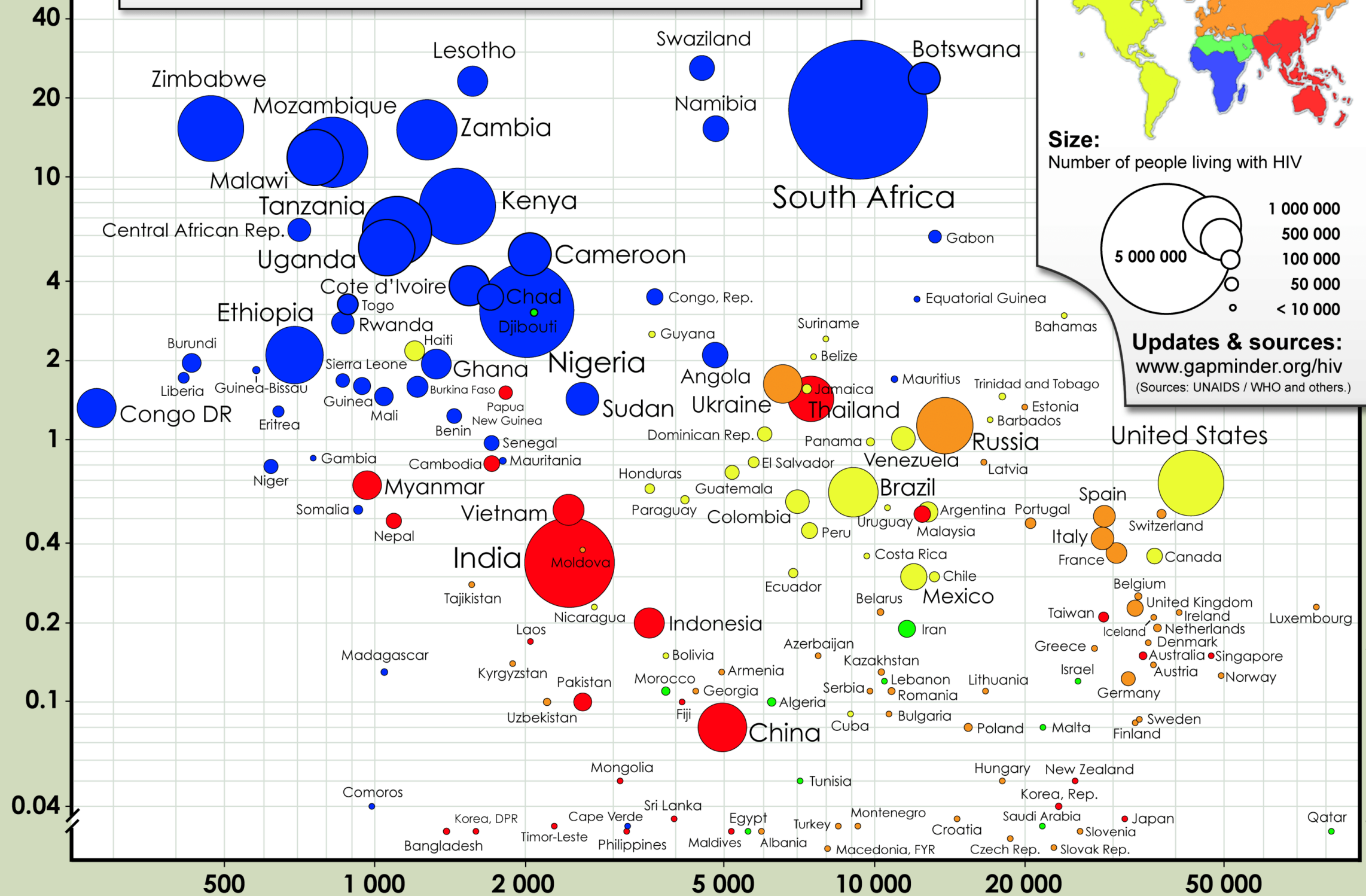
Video source: [Insights on HIV, in stunning data visuals](#)

Insights on HIV, in stunning data visuals

Hans Rosling
Insights on HIV, in stunning data visuals

HIV infected (% of adults aged 15-49 years)

Gapminder HIV Chart 2009 (Data from 2007)



Income per person (\$ per year in purchasing power parity)

Free to redistribute,
PDF available at
www.gapminder.org/hiv



Key Messages:

- No simple answer to the complex question.
- There are no two types of countries one with and another without the HIV pandemic.
- To see more, visit:
 - gapminder.org
 - [Ted talks of Hans Rosling](#)

Other homework

At home: Do exercises from the lecture

At home: Readings

- ([Kuchenbecker et al. 2017](#)) **before the week 4;**
- Chapter 1 in “The Book of Why: The New Science of Cause and Effect” by Judea Pearl and Dana Mackenzie

At home: Explore the best use of a scatter plots with **Gapminder**

Play with data using gap minder:

- <https://www.gapminder.org/tools/>

Watch some Hans Rosling's Ted Talks:

- https://www.youtube.com/playlist?list=PLSrgd_ElrrXegdtDHPmUzrzLpwZ2dSYKg

Explore the gap-minder foundation channel

- <https://www.youtube.com/channel/UCMbmqUMqzerMSM48pGYMYsg>

Key concepts covered in the lecture

Scatter plot:

- How to construct it;
- Linear/non-linear relation between variables
- Variability.

Trend line:

- Linear trend;

Correlation:

- Linear relationship only.
- Interpretation Positive/Negative coefficient
- Non-linearity, outliers and correlation.
- Correlation is not causality.

The Ceteris Paribus:

- Ceteris Paribus;
- Counterfactual and potential outcome;
- Cause and treatment;
- Outcome and Effect;

References

- Diez, David, Mine Cetinkaya-Rundel, and Christopher D Barr. 2022. *OpenIntro Statistics: Fourth Edition*. OpenIntro. <https://www.openintro.org/book/os/>.
- Gerbrandy, J. L., and E. A. van Dura. 1972. “Anamnestic Secretory Antibody Response in Respiratory Secretions of Intranasally Immunized Mice.” *Journal of Immunology (Baltimore, Md. : 1950)* 109 (November): 1146–48.
- Kuchenbecker, Judith, Anika Reinbott, Beatrice Mtimuni, Michael B. Krawinkel, and Irmgard Jordan. 2017. “Nutrition Education Improves Dietary Diversity of Children 6-23 Months at Community-Level: Results from a Cluster Randomized Controlled Trial in Malawi.” Edited by Jacobus P. van Wouwe. *PLOS ONE* 12 (4): e0175216. <https://doi.org/10.1371/journal.pone.0175216>.
- Wooldridge, M. Jeffrey. 2020. *Introductory Econometrics: A Modern Approach*. South-Western. <https://www.cengage.uk/shop/isbn/9781337558860>.

